A

Project Report

On

"A MACHINE LEARNING SYSTEM FOR INDOOR HAZARDOUS GAS MONITORING AND PREDICTION IN REMOTE AREAS"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES

(Affiliated to J.N.T.U. Anantapur, Approved by A.I.C.T.E, New Delhi)

Accredited by NAAC with 'A' Grade, Bangalore & NBA (EEE. ECE, CSE)

Utukur (P), C.K. Dinne (V&M), Kadapa-516003

ANDHRA PRADESH

2025-2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project work titled **"A MACHINE LEARNING SYSTEM FOR INDOOR HAZARDOUS GAS MONITORING AND PREDICTION IN REMOTE AREAS"** is a bonafide work done by

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DECLARATION

We hereby declare that the project report entitled as **"A MACHINE LEARNING SYSTEM FOR INDOOR HAZARDOUS GAS MONITORING AND PREDICTION IN REMOTE AREAS"** is a record of project work carried out by us and submitted to Jawaharlal Nehru Technological University Ananthapur in partial fulfilment for award of the degree of Bachelor of Technology in Computer Science and Engineering. The work carried out is original and has not been submitted to any other Universities or Institutions for the award of any credits.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modem engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

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PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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PO10: Communication: Communicate effectively on complex engineering activities with the Engineering community and with society at large, such as, being able to

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PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

PROGRAM SPECIFIC OUTCOMES (PSO'S)

PSO-1: Ability to learn the applicability of various systems software elements for solving design problems.

PSO-2: Ability to apply their skills in the field of Web Designing, Cloud Computing, Machine Learning, Artificial Intelligence and Internet of Things.

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Mapping of PO's and PSO's through Project

Title of the Project	A MACHINE LEARNING SYSTEM FOR INDOOR HAZARDOUS GAS MONITORING AND PREDICTION IN REMOTE AREAS	
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SYNOPSIS

This project focuses on developing a machine learning-based system for indoor hazardous gas monitoring and prediction in remote areas. It uses sensors to collect real-time data such as gas concentration, temperature, and humidity. The collected data is analyzed using machine learning algorithms to detect and predict dangerous gas levels. The system provides early warnings and remote monitoring to improve safety and reduce risks. Overall, it offers a cost-effective and efficient solution for preventing accidents in indoor environments. It is designed to be reliable, scalable, and suitable for various real-world applications.

Title of the Project	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2
A Machine Learning System For Indoor Hazardous Gas Monitoring And Prediction In Remote Areas	3	3	3	3	3	1	1	1	2	2	1	2	3	3

ABSTRACT

The machine learning-based system for indoor hazardous gas monitoring and prediction in remote areas. The system utilizes gas sensors to collect real-time environmental data, including gas concentration, temperature, and humidity. The collected data is processed using machine learning algorithms to detect abnormal patterns and predict potential hazardous situations. The proposed approach enables early warning through timely alerts, reducing risks to human health and property. Additionally, the system supports remote monitoring, making it suitable for isolated and low-resource environments. Experimental results demonstrate improved accuracy and reliability in gas detection and prediction compared to traditional monitoring methods. Overall, this approach enhances safety, reduces risk, and provides a cost-effective solution for hazardous gas monitoring in remote indoor environments.

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LIST OF ABBREVIATION

Abbreviation	Description
ML	Machine Learning
SVM	Support Vector Machine
BI-LSTM	Bi-Directional Long Short-Term Memory
RFM	Recency, Frequency, Monetary
API	Application Programming Interface
DB	Database
UI	User Interface
URL	Uniform Resource Locator
CSV	Comma-Separated Values
HTTP	Hypertext Transfer Protocol
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets

1. INTRODUCTION

1.1 INTRODUCTION

Indoor air quality plays a vital role in maintaining human health and safety, especially in environments where hazardous gases may be present. In remote areas such as industrial sites, storage units, mines, and isolated buildings, monitoring gas levels becomes challenging due to limited access and lack of continuous supervision. Harmful gases like carbon monoxide, methane, and LPG can lead to serious health risks, fire hazards, or even explosions if not detected early. Traditional gas monitoring systems mainly focus on detection and alarm generation after gas levels exceed a certain threshold. However, these systems do not provide predictive analysis or early warnings before the situation becomes critical. This limitation increases the risk of accidents, particularly in remote locations where immediate response is not always possible.

To address these challenges, this project proposes a machine learning-based system for indoor hazardous gas monitoring and prediction. The system uses sensors to continuously collect environmental data such as gas concentration, temperature, and humidity. Machine learning algorithms are then applied to analyze the data, identify patterns, and predict potential gas hazards in advance. The integration of Internet of Things (IoT) technology enables real-time data transmission and remote monitoring, allowing users to access information from anywhere. This makes the system more efficient, reliable, and suitable for remote applications. By combining sensing, data analysis, and prediction, the proposed system aims to improve safety, reduce risks, and provide a smart solution for hazardous gas monitoring.

The system is designed to be scalable and adaptable to different environments by adjusting sensor types and machine learning models. It can be implemented with low-cost hardware, making it accessible for widespread use. The use of automated alerts such as SMS or app notifications ensures quick response in emergency situations. Furthermore, continuous data collection helps in improving model performance over time, making predictions more accurate and reliable. This also supports long-term monitoring and data-driven decision-making for safety management. The system can be integrated with cloud platforms for secure data storage and remote accessibility. It also enables real-time analytics and visualization, enhancing user awareness and efficient system control.

1.2 PROJECT OBJECTIVES

- The main objective of this project is to design and develop a machine learning-based system for monitoring and predicting hazardous gases in indoor environments, especially in remote areas. The system aims to continuously collect environmental data such as gas concentration, temperature, and humidity using sensors, and analyze this data using machine learning algorithms. This helps in detecting abnormal gas levels and predicting potential hazards before they become dangerous.
- Another important objective is to provide real-time alerts and remote monitoring capabilities to improve safety and reduce risks. The system focuses on accuracy, reliability, and early warning to prevent accidents such as gas leaks, fires, or health issues. Additionally, it aims to create a low-cost, energy-efficient, and scalable solution that can be easily implemented in various remote locations.

1.3 SCOPE OF THE PROJECT

- The scope of this project includes the development and implementation of a smart gas monitoring system that integrates sensors, machine learning, and IoT technology. It covers data collection, data processing, model training, and real-time prediction of hazardous gas conditions. The system can be used in indoor environments such as industries, warehouses, laboratories, and remote buildings where continuous monitoring is essential.
- The project also aims to enhance decision-making by providing data visualization through dashboards and reports. This enables users to monitor trends, analyze historical data, and take preventive actions. It supports continuous learning by updating models with new data to improve prediction accuracy over time.
- Furthermore, the project can be extended to support multiple gas types and improved prediction models for higher accuracy. It also allows future enhancements such as mobile application integration, cloud data storage, and advanced analytics for better deSScision-making. The system can be scaled for large environments and adapted to different use cases, making it a flexible and practical solution for safety management.

2. SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

- Traditional gas monitoring systems mainly use basic gas sensors to detect the presence of hazardous gases.
- These systems work based on fixed threshold values and trigger alarms only after gas levels become dangerous.
- They do not provide prediction or early warning before the hazard occurs. Most systems require manual monitoring and lack automation in data analysis.
- Limited or no support for remote monitoring, especially in isolated or rural areas. Data collected is often not stored or used for future analysis and improvement.
- These systems have lower accuracy and efficiency compared to advanced machine learning-based approaches. They are not capable of adapting to changing environmental conditions.
- Integration with modern technologies like IoT and cloud platforms is very limited. Maintenance and calibration of sensors can be difficult and time-consuming.

2.1.1 DISADVANTAGES OF EXISTING SYSTEM

- Traditional gas monitoring systems rely on fixed threshold values, which means they can only detect hazardous gases after the concentration reaches a dangerous level, lacking the ability to predict or prevent potential hazards in advance.
- These systems often generate delayed alerts, which can increase the risk of accidents, health issues, or damage to property, especially in critical environments where immediate action is required.
- Most existing systems do not support remote monitoring, making them ineffective for use in remote or isolated areas where continuous human supervision is not possible.
- Existing systems require frequent maintenance and calibration of sensors, which can be time-consuming, costly, and difficult to manage, particularly in large-scale or remote deployments.
- They lack advanced data analytics capabilities, limiting insights for proactive decision-making and long-term safety improvements.

2.2 PROPOSED SYSTEM

- The proposed system uses advanced and sensitive gas sensors to continuously monitor indoor environmental conditions and detect the presence of hazardous gases in real time with improved reliability.
- It collects real-time environmental data such as gas concentration levels, temperature, and humidity, ensuring continuous observation of the surroundings without interruption.
- Machine learning algorithms are applied to analyze the data and identify patterns. The system predicts hazardous gas conditions before they become critical.
- It provides early alerts through notifications such as alarms, SMS, or mobile apps. ensuring quick response even in remote areas.
- Data is stored for future analysis and model improvement.
- The system supports multi-gas detection for better safety coverage. It uses cloud platforms for secure data storage and easy access.
- The system is designed with low power consumption for long-term operation.

2.2.1 ADVANTAGES OF PROPOSED SYSTEM

- The system provides early prediction of hazardous gas conditions, reducing the chances of accidents, health risks, and environmental damage.
- The system is cost-effective, scalable, and easily integrates with modern technologies like IoT and cloud computing, while also supporting continuous improvement through data storage and analysis.
- Remote monitoring capability makes it highly suitable for isolated and hard-to-reach areas, reducing the need for constant human supervision.
- High accuracy and reliability are achieved through machine learning-based data analysis, outperforming traditional threshold-based systems.
- Automation reduces manual effort and supports real-time decision-making, making the system efficient and user-friendly.
- The system offers real-time data visualization through dashboards, enabling users to monitor trends, analyze patterns, and make informed safety decisions quickly and effectively.

3. SYSTEM REQUIREMENTS AND SPECIFICATIONS

3.1 HARDWARE REQUIREMENTS

- Processor : Core i3
- RAM : 4 GB (min)
- Hard Disk : 200 GB
- Key Board : Standard Windows Keyboard
- Mouse : Two or Three Button Mouse
- Monitor : SVGA

3.2 SOFTWARE REQUIREMENTS

- Operating system : Windows 7 or higher.
- Coding Language : Python.
- Front-End : Html5, CSS3, JavaScript
- Back-End : Django
- Domain : Machine Learning

3.3 SOFTWARE DESCRIPTION

The software system is developed using Python to process real-time sensor data such as gas levels, temperature, and humidity, and applies machine learning to predict hazardous conditions. It enables remote monitoring, data storage, and alert notifications for improved safety.

3.3.1 PYTHON

Python is a high-level, interpreted programming language widely used for software development, data analysis, and machine learning applications. It is known for its simple and readable syntax, which makes it easy for beginners to learn and use. Python supports multiple programming paradigms such as object-oriented, procedural, and functional programming. It provides powerful libraries like Pandas, NumPy, and Scikit-learn for efficient data handling and model building. Python is also widely used in developing data-driven systems due to its flexibility and scalability. Its simplicity and strong ecosystem make it one of the most preferred programming languages today.

- **Python is Interpreted:** Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive:** You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented:** Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language:** Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Additionally, Python supports rapid prototyping and quick application development. It has a vast ecosystem of third-party libraries that simplify complex tasks. Python is widely adopted in academic, industrial, and research domains. Its versatility allows it to be used in various fields such as automation, web development, and artificial intelligence.

Python also emphasizes code simplicity and maintainability, making it ideal for long-term software development. Its clear and readable syntax allows developers to write programs with fewer lines of code. This not only improves productivity but also

reduces the chances of errors during development. As a result, Python is widely preferred for building efficient and scalable applications.

Table 3.1 Python Implementation Details

S.No	Component	Description
1	Data Handling	Used Pandas library to load, clean, and manage dataset
2	Numerical Processing	NumPy used for mathematical operations and array handling
3	Data Preprocessing	Performed cleaning, encoding, and normalization of data
4	Feature Engineering	Extracted features like spending, frequency, and patterns
5	Model Building	Implemented Decision Tree and Random Forest algorithms
6	Model Training	Trained model using processed dataset for prediction
7	Model Storage	Saved trained model using Pickle as model.pkl
8	Data Visualization	Used Matplotlib and Seaborn to display graphs and insights
9	Model Evaluation	Evaluated model using accuracy, precision, recall, F1 score
10	Prediction System	Generated customer segment predictions (High/Medium/Low)
11	Dashboard Integration	Displayed results using charts and user-friendly interface

3.3.2 HISTORY OF PYTHON

Python was developed by Guido van Rossum and was first released in 1991. It was created with the aim of making programming simple, readable, and efficient. The name "Python" was inspired by the popular British comedy show Monty Python's Flying Circus. Over time, Python evolved through various versions, with Python 3 becoming the standard version due to its improved features. It gained popularity rapidly due to its ease of use and powerful capabilities. Today, Python is widely used in areas such as web development, data science, artificial intelligence, and automation.

Over the years, Python has received continuous updates to improve performance and usability. The language community actively contributes to its development and enhancement. Python's growing popularity has made it one of the most in-demand programming languages. Its adaptability ensures its relevance in modern technological advancements.

Python was designed to be highly readable, using English keywords where others use punctuation, and it enforces a, philosophy often summarized in the "Zen of Python" Python was designed to be highly readable, using English keywords where others use punctuation, and it enforces a, philosophy often summarized in the "Zen of Python"

Python has a strong and active global community that continuously contributes to its growth and improvement. Developers around the world create libraries, frameworks, and tools that extend Python's capabilities. Popular frameworks like Django and Flask have made Python a preferred choice for web development. Community support through forums, documentation, and open-source contributions makes problem-solving easier. This widespread adoption has strengthened Python's position in both academia and industry.

Python is also widely used in the field of data science and machine learning due to its powerful libraries and simplicity. Libraries such as Pandas, NumPy, and Scikit-learn provide efficient tools for data analysis and model building. Python enables developers to handle large datasets and perform complex computations with ease. Its integration with visualization tools helps in better understanding of data patterns. This makes Python highly suitable for data-driven applications and research. It also supports advanced frameworks like TensorFlow and PyTorch for deep learning applications. This further enhances its capability in building intelligent and predictive systems.

3.3.3 FEATURES OF PYTHON

Python is a programming language known for its simplicity and readability. It uses a clean and easy-to-understand syntax, making it suitable for beginners as well as experienced developers. Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming. It is dynamically typed, meaning variable types are determined at runtime, which increases flexibility in coding. Python is platform-independent, allowing programs to run on different operating systems without modification. It offers a vast standard library and external libraries for tasks such as data analysis, machine learning, web development, and automation. Being open-source, Python is free to use and continuously improved by a large global community. Overall, Python is widely used due to its versatility, efficiency, and strong support for modern technologies like AI and data science.

The python programming language is a high-level language that can be characterised by all the following features (or) buzzwords.

1. **SIMPLE AND READABLE SYNTAX:**

   Python has a clean and easy-to-understand syntax that resembles natural language. It reduces the complexity of coding and improves readability. This makes it ideal for beginners and quick development. It also enhances collaboration among developers by making code easier to understand and maintain. This results in faster development cycles and improved software quality.

2. **INTERPRETED LANGUAGE:**

   Python is executed line by line without the need for compilation. This makes debugging easier and faster. It also allows rapid testing and development. Errors can be identified and fixed immediately during execution. This improves development efficiency and reduces overall coding time.

3. **HIGH-LEVEL PROGRAMMING:**

   Python abstracts low-level details like memory management. Developers can focus on logic instead of system-level operations. This improves productivity and reduces coding effort. It automatically manages memory using built-in garbage collection. This reduces the chances of memory leaks and related errors. It also simplifies program development and enhances overall code efficiency.

4. OBJECT-ORIENTED PROGRAMMING (OOP):

Python supports OOP concepts like classes, objects, inheritance, and polymorphism. It helps in organizing code into reusable components. This makes programs modular and scalable.

5. DYNAMIC TYPING:

Python does not require explicit declaration of variable types. The type is determined at runtime automatically. This provides flexibility and speeds up development. It allows developers to write less code and focus on problem-solving. This dynamic nature makes prototyping faster and more efficient. It also simplifies code maintenance and reduces complexity.

6. CROSS-PLATFORM COMPATIBILITY:

Python provides a rich set of libraries for various applications. NumPy is used for numerical operations, Pandas for data handling, and Scikit-learn for machine learning. These libraries save time and effort in development.

7. EXTENSIVE LIBRARIES:

Python provides a rich set of libraries for various applications. NumPy is used for numerical operations, Pandas for data handling, and Scikit-learn for machine learning. These libraries save time and effort in development. They offer pre-built functions and tools that simplify complex tasks. This helps developers build efficient and high-performance applications quickly.

8. MACHINE LEARNING & AI SUPPORT:

Python is widely used in machine learning and artificial intelligence. It provides frameworks like TensorFlow and PyTorch. This enables efficient model building and deployment.

9. DATA ANALYSIS AND VISUALIZATION:

Python supports powerful tools for analyzing and visualizing data. Libraries like Matplotlib and Seaborn help create graphs and charts. This makes data interpretation easier. It enables users to identify patterns, trends, and insights effectively. This improves decision-making and enhances data-driven analysis.

10. AUTOMATION AND SCRIPTING:

Python is commonly used for automating repetitive tasks. It can handle file operations, web scraping, and system tasks. This increases efficiency and reduces manual work. It also minimizes human errors and ensures consistency in task execution. This allows developers to focus on more complex and important problems.

11. OPEN SOURCE:

Python is free to use and distribute. Its source code is publicly available for modification. This encourages innovation and community contributions. It also allows developers to customize the language according to their needs.

12. LARGE COMMUNITY SUPPORT:

Python has a vast and active developer community. It provides extensive documentation and online support. This helps in quick problem-solving and learning.

3.3.4 ADVANTAGES OF PYTHON

- Easy to read and write programs
- Faster development due to simple syntax
- Wide range of libraries for different applications
- Highly suitable for machine learning and data analysis
- Open-source and freely available
- Flexible and scalable for large applications
- Strong community support and resources
- Easy integration with other programming languages
- Supports rapid application development

3.3.5 MODULES OF PYTHON

A. PANDAS

Pandas is a powerful open-source library used for data manipulation and analysis. It provides data structures such as Data-Frames and Series, which make it easy to handle structured data. It is widely used for cleaning, filtering, and transforming datasets. Pandas supports reading data from various file formats like CSV, Excel, and databases. It also allows efficient handling of missing values and data aggregation. This makes it highly useful for preparing data before applying machine learning algorithms.

A. NUMPY

NumPy is a fundamental library for numerical computing in Python. It provides support for multi-dimensional arrays and matrices, along with a wide range of mathematical functions. NumPy enables fast computation and efficient storage of numerical data. It is commonly used for performing operations such as sorting, reshaping, and mathematical calculations. The library is highly optimized for performance and is widely used in scientific computing. It also serves as a foundation for many other libraries like Pandas and Scikit-learn.

B. SCIKIT-LEARN

Scikit-learn is a popular machine learning library used for building and evaluating models. It provides various algorithms such as Decision Tree, Random Forest, and many others. The library supports tasks like classification, regression, and clustering. It also includes tools for model training, testing, and performance evaluation. Scikit-learn simplifies the implementation of complex machine learning techniques. It is widely used due to its efficiency, simplicity, and integration with other Python libraries.

C. MATPLOTLIB

Matplotlib is a widely used library for data visualization in Python. It helps in creating graphs such as line charts, bar graphs, histograms, and scatter plots. The library provides flexibility in customizing visual elements like labels, colours, and styles. It is useful for presenting data in a clear and understandable manner. Matplotlib plays a key role in analyzing trends and patterns visually.

D. ENSORFLOW

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google. TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015. It provides a flexible architecture that allows deployment across CPUs, GPUs, and TPUs. TensorFlow supports deep learning, computer vision, and natural language processing applications.

E. JOBLIB

Joblib is a Python library used for efficiently saving and loading machine learning models and large data objects. It is commonly used with libraries like Scikit-learn to persist trained models to disk. Joblib is optimized for handling large NumPy arrays and supports fast serialization using compression techniques. It also provides simple utilities for parallel computing to speed up tasks. Overall, Joblib helps in improving performance and reusability in machine learning workflows. It is especially useful in deployment scenarios where trained models need to be reused without retraining. Additionally, it reduces memory usage and speeds up data processing in large-scale applications.

3.4 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

1. ECONOMICAL FEASIBILITY
2. TECHNICAL FEASIBILITY
3. SOCIAL FEASIBILITY

3.4.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus, the developed system is well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased. The use of open-source tools and platforms further reduces overall development and maintenance costs. Additionally, the system minimizes operational expenses by automating monitoring and alert processes. Hence, the solution is cost-effective and suitable for deployment even in resource-constrained environments.

3.4.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources and may burden the client. The developed system must have a modest requirement, as only minimal or no changes are required for implementing this system. The system is designed using scalable and flexible technologies that support easy integration with existing infrastructure. It also ensures compatibility with standard hardware and IoT devices used for gas monitoring. Furthermore, the system is reliable and can operate efficiently even in low-network or remote area conditions. It also supports easy maintenance and upgrades without affecting overall system performance.

3.4.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system. The system is designed with a user-friendly interface to ensure ease of use for individuals with minimal technical knowledge. Proper training sessions and guidance materials will be provided to improve user understanding and comfort.

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The proposed system architecture is a multi-layered design combining sensor networks, edge computing, cloud services, and user interfaces. At the sensor layer, distributed low-cost gas sensors (such as MQ-2, MQ-7, and electrochemical sensors) are deployed across target areas to detect gases like methane, carbon monoxide, ammonia, and hydrogen sulfide. These sensors are paired with environmental sensors for temperature, humidity, and wind direction to contextualize gas readings. Data from the sensors is transmitted via low-power wireless communication protocols (e.g., LoRaWAN, NB-IoT) to local edge devices. The edge layer performs immediate preprocessing, such as noise filtering, outlier detection, and basic anomaly identification, reducing the volume of data transmitted to the cloud. At the cloud layer, machine learning models—including time-series forecasting (BI-LSTM, SVM), ensemble regressors, and anomaly detection algorithms—process the aggregated data to predict hazardous gas trends, assess risks, and generate actionable alerts. A user interface layer provides dashboards and notifications accessible via web and mobile applications, showing real-time gas levels, predictive trends, and historical analysis. Security measures, including encryption, authentication, and data integrity checks, are embedded throughout the architecture to ensure safe transmission and storage. This layered approach ensures reliability, scalability, and resilience, enabling deployment in remote locations where continuous monitoring is critical for environmental safety and public health. The system supports real-time data synchronization to ensure up-to-date monitoring across all connected devices. It can be expanded to include more sensors and advanced models as needed, making it highly flexible for different environments. The architecture also enables automated decision-making, reducing human intervention and improving overall system efficiency. Regular model updates and continuous learning techniques are used to improve prediction accuracy and adapt to changing environmental conditions over time. Additionally, the system supports integration with emergency response services for faster incident handling. It also enables predictive maintenance of sensors by identifying performance degradation in advance. Furthermore, energy-efficient operation ensures longer device lifespan and reduced operational costs.

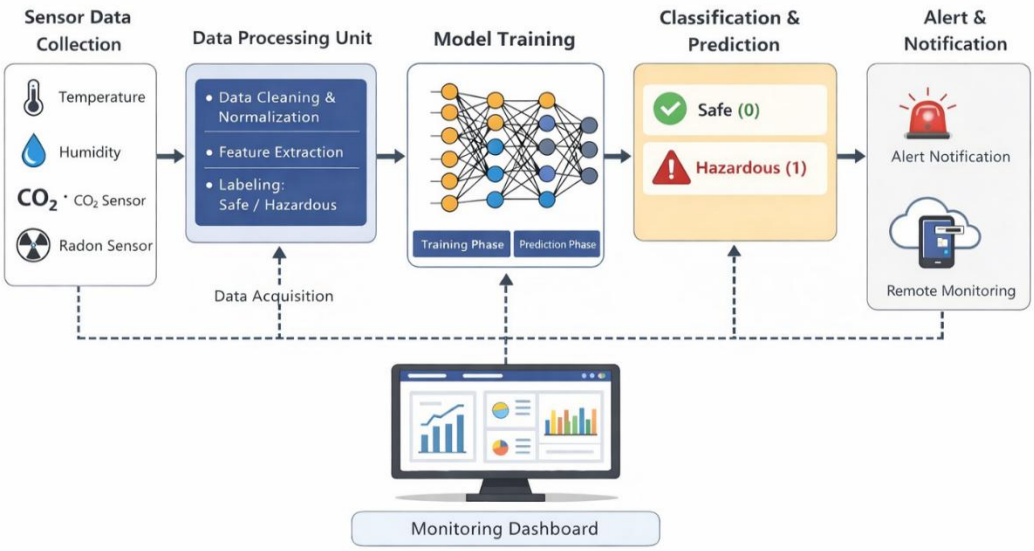


Figure 4.1 System architecture

The architecture represents an IoT-based air quality monitoring and hazard prediction system. It begins with the Sensor Data Collection layer, where sensors measure parameters like temperature, humidity, CO_2, and radon levels. This data is sent to the Data Processing Unit, where it undergoes cleaning, normalization, feature extraction, and labeling as safe or hazardous. The processed data is then used in the Model Training phase, where machine learning models learn patterns during training and generate predictions during deployment. In the Classification & Prediction stage, the system classifies the environment as safe or hazardous based on learned patterns. If a hazardous condition is detected, the Alert & Notification module triggers warnings and enables remote monitoring. All system data and predictions are visualized through a Monitoring Dashboard, allowing users to track real-time conditions and historical trends efficiently. Additionally, the system supports continuous data acquisition and model updates to improve prediction accuracy over time. It also ensures scalability and integration with cloud platforms for efficient storage, processing, and remote accessibility.

4.2 MODULES DESCRIPTION

The module description explains how the entire system is divided into smaller functional units, each performing a specific task such as data collection, processing, prediction, and alert generation. It helps in understanding the working flow of the system in a structured and organized manner. This approach improves system design, implementation, and maintenance, especially for complex applications like hazardous gas monitoring in remote areas.

It enhances system flexibility by allowing each module to be developed and tested independently. It also supports scalability, making it easier to upgrade or add new features without affecting the entire system. This modular approach ensures better reliability and efficient performance of the overall system.

MODULES

- Sensor Data Collection Module
- Data Preprocessing unit Module
- Model Training Module
- Classification & Prediction Module
- Alert & Notification Module
- Monitoring Dashboard Module

4.2.1 SENSOR DATA COLLECTION MODULE

In this module, it is responsible for collecting real-time data from various sensors such as gas sensors (MQ-2, MQ-7), temperature, and humidity sensors. It continuously monitors environmental conditions and sends the collected data to the next stage for processing.

4.2.2 DATA PREPROCESSING UNIT MODULE

In this module, it cleans and prepares the collected data for analysis. It includes processes such as data cleaning, normalization, and feature extraction. It also labels the data (safe or hazardous) to improve the accuracy of the machine learning model. Additionally, this module removes inconsistencies and ensures that the data is in a structured format suitable for processing. It plays a crucial role in enhancing model performance by providing high-quality and reliable input data.

4.2.3 MODEL TRAINING MODULE

In this module, it uses algorithms such as Bi-LSTM, GRU, SVM or other models to analyze the processed data. It identifies patterns and predicts hazardous gas conditions before they become critical, enabling early warning. The models are trained on historical data to improve prediction accuracy and reliability. This module also continuously updates predictions based on real-time data inputs for dynamic decision-making.

4.2.4 CLASSIFICATION AND PREDICTION MODULE

In this module, it classifies the gas conditions into categories like safe or hazardous based on the model output. After processing the data, this module classifies the environment into safe or hazardous categories. It also provides predictive outputs indicating whether gas levels are likely to increase or become dangerous. This helps in taking preventive actions in advance and improves system reliability.

4.2.5 ALERT AND NOTIFICATION MODULE

In this module, the system detects or predicts hazardous gas levels, this module generates alerts to notify users. Alerts can be in the form of alarms, SMS, or mobile app notifications. This ensures immediate awareness and allows quick response to prevent accidents or health risks. Additionally, the module can be configured with threshold values to trigger alerts at different risk levels. It also maintains a log of alerts for future analysis and system improvement.

4.2.6 MONITORING DASHBOARD MODULE

In this module, it provides a user-friendly dashboard to display real-time data. It displays current gas levels, predictions, and historical data through graphs and dashboards. Users can access this information remotely, making it easier to manage and analyze the system efficiently. Additionally, the dashboard supports data visualization tools such as charts and alerts for better understanding of trends. It also enables users to download reports and monitor system performance over time for improved decision-making.

4.3 INTRODUCTION TO UML

Unified Modelling Language (UML) is a standardized visual modelling language used to design, represent, and document the structure and behaviour of a system. In the project. UML helps in clearly illustrating system components, data flow, and interactions between different modules. The Object Management Group (OMG) adopted the UML as a standard in 1997. International Organization for Standardization (ISO) published UML in 2005, It is reviewed periodically.

UML makes it easier for developers, designers, and stakeholders to communicate and understand the system architecture and functionality. It also helps in planning and improving the system before actual implementation.

Why we have to use UML in projects?

- Complex applications need collaboration and planning from multiple teams and hence require a clear and concise way to communicate amongst them.
- Businessmen do not understand code. So, UML becomes essential to communicate with non-programmers, essential requirements, functionalities and process of the system.
- A lot of time is saved down the line when the teams are able to visualize processes, user interactions and static structure of the system.
- UML is linked with objected-oriented design and analysis. UML makes the use of elements and forms associations between them to form diagrams. Diagrams in UML can be broadly classified as

In the project four basic UML diagrams have been explained

1. Class Diagram
2. Use Diagram
3. Sequence Diagram
4. Collaboration Diagram
5. Deployment Diagram
6. State chart Diagram
7. Component Diagram
8. Activity Diagram

4.3.1 CLASS DIAGRAM

A UML Class Diagram is a type of structural diagram that represents the static structure of a system by showing its classes, attributes, methods, and the relationships between them. It is mainly used to model the object-oriented design of a system and helps in understanding how different parts of the system are organized and connected.

In this diagram, each class is represented as a box divided into three sections: the class name, attributes (data members), and methods (functions). Relationships such as association, inheritance, aggregation, and composition are shown using connecting lines and arrows. These relationships define how classes interact with each other.

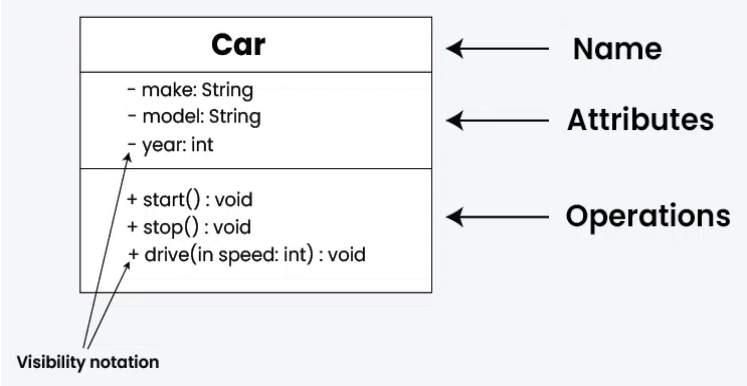


Figure 4.2 Class Diagram

- The top compartment contains the name of the class. It is printed in bold and centered, and the first letter is capitalized.
- The middle compartment contains the attributes of the class. They are left-aligned and the first letter is lower case
- The bottom compartment contains the operations the class can execute. They are also left-aligned and the first letter is lower case
- In the design of system, a number of classes are identified and grouped together in a class diagram the classes of conceptual design are often split into subclasses.

4.3.2 USE CASE DIAGRAM

A Use Case Diagram is a behavioural diagram that represents the functional requirements of a system by showing how users (actors) interact with the system. It provides a simple and clear view of what the system does without going into technical details. This diagram helps in identifying different users and the actions they can perform within the system.

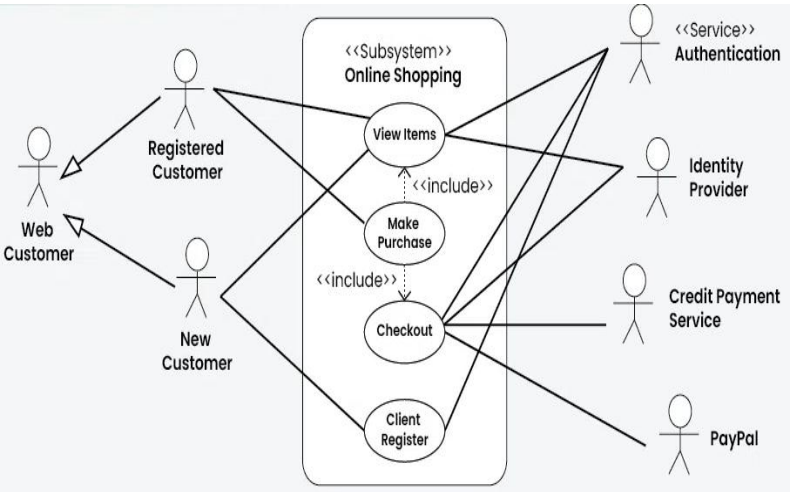


Figure 4.3: Use Case Diagram

4.3.3 SEQUENCE DIAGRAM

A Sequence Diagram is a behavioural diagram that shows how different components of a system interact with each other in a particular sequence over time. It focuses on the order of messages exchanged between objects or modules to complete a specific task. This diagram helps in understanding the dynamic behaviour of the system.

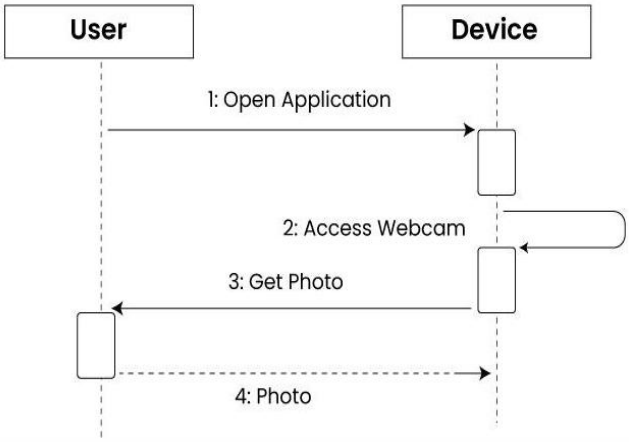


Figure 4.4 Sequence Diagram

4.3.4 COLLABORATION DIAGRAM

A Collaboration Diagram (also known as a Communication Diagram) shows how different objects in a system interact with each other through messages to complete a task, focusing on their relationships and communication structure. It helps in understanding how system components are connected and how they work together to achieve functionality.

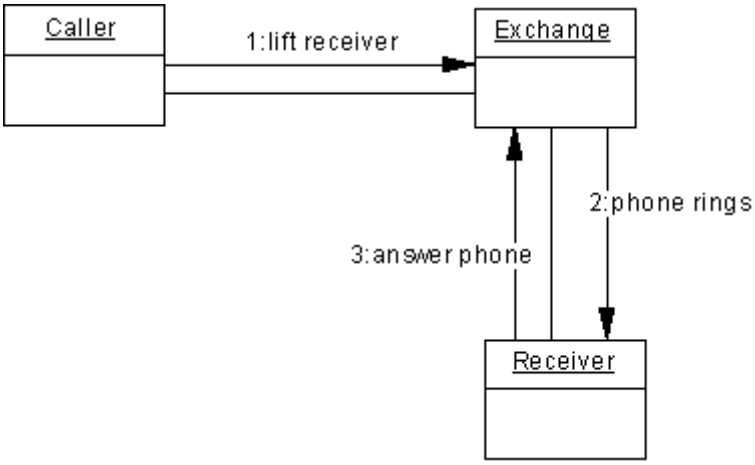


Figure 4.5: Collaboration Diagram

4.3.5 DEPLOYMENT DIAGRAM

A Deployment Diagram is a type of Structural UML Diagram that shows the physical deployment of software components on hardware nodes. It illustrates the mapping of software components onto the physical resources of a system, such as servers, processors, storage devices, and network infrastructure.

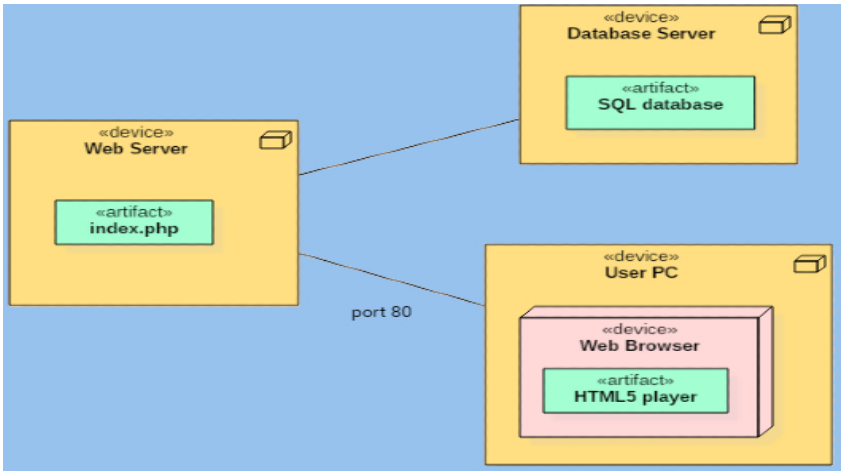


Figure 4.6 Deployment Diagram

4.3.6 STATE CHART DIAGRAM

State chart diagram describes the flow of control from one state to another state. States are defined as a condition in which an object exists and it changes when some event is triggered. The most important purpose of State chart diagram is to model lifetime of an object from creation to termination.

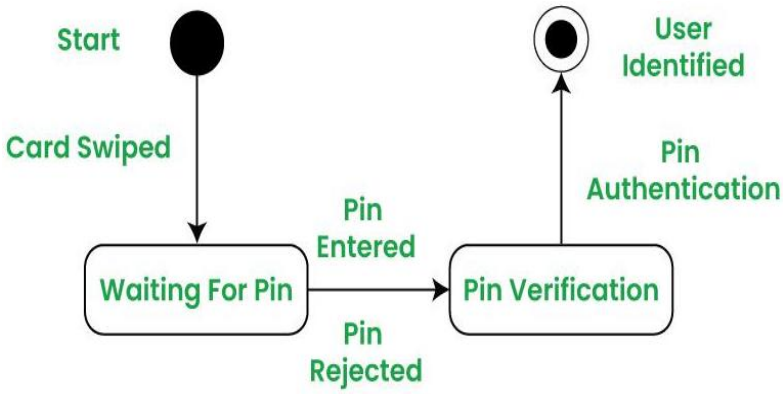


Figure 4.7 State Chart Diagram

4.3.7 COMPONENT DIAGRAM

A Component Diagram represents the high-level structure of a system by showing its components and how they are connected or interact with each other. It helps in understanding system organization, dependencies, and how different parts work together to deliver functionality.

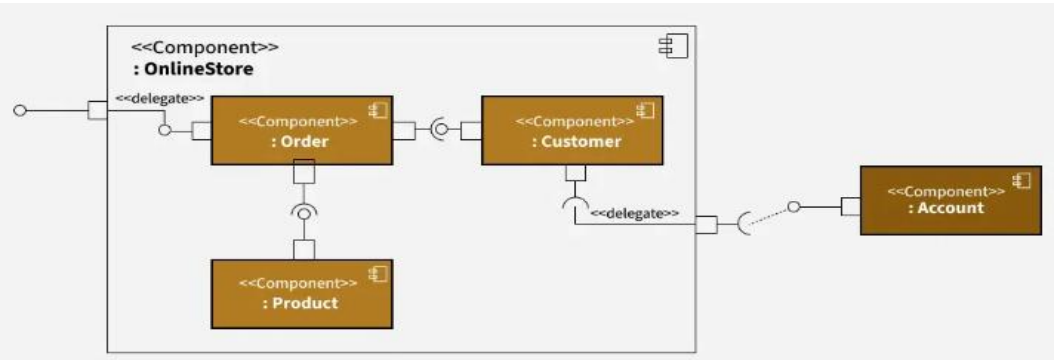


Figure 4.8 Component Diagram

4.3.8 ACTIVITY DIAGRAM

The Activity Diagram represents the workflow of a system by showing the sequence of activities and decision points involved in a process. It helps in understanding the step-by-step execution and control flow of operations in the system.

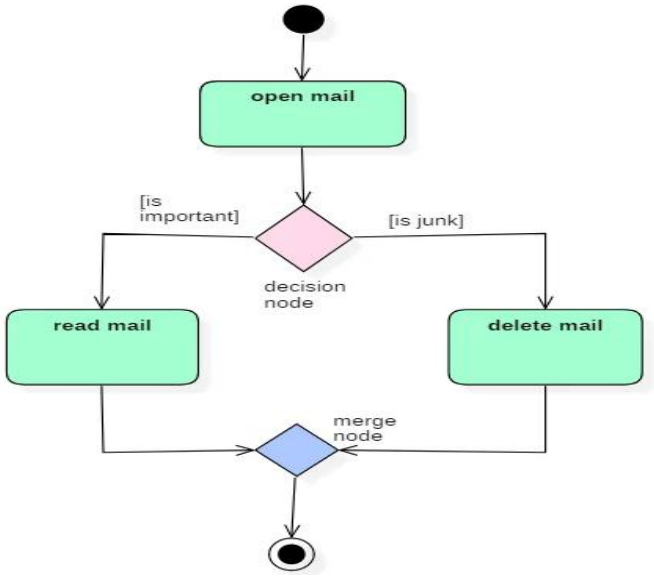


Figure 4.9 Activity Diagram

4.4 DATA FLOW DIAGRAM

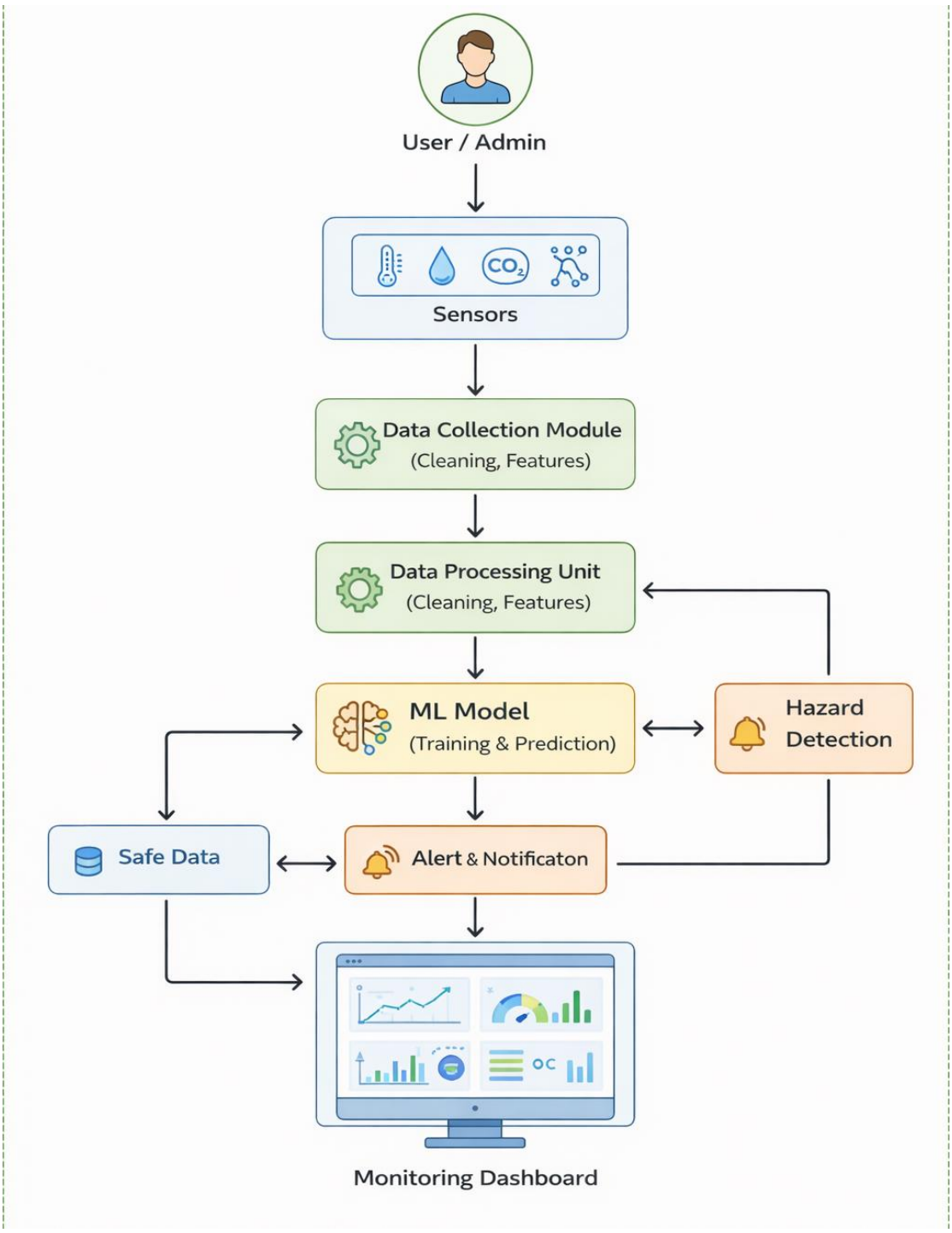


Figure 4.10 Data Flow Diagram

4.5 CLASS DIAGRAM FOR OVERALL PROJECT

The class diagram represents a machine learning–based system for indoor hazardous gas monitoring and prediction. The system begins with Sensor Data Collector, which gathers environmental data such as temperature, humidity, CO_2, and radon levels from sensors. This raw data is passed to the Processing Unit, where it is cleaned, normalized, and converted into meaningful features for analysis. The processed data is then used by the Bi-LSTM Model Trainer to build, train, and evaluate a machine learning model that predicts hazardous conditions. The Hazard Detector receives new sensor inputs, applies preprocessing, and uses the trained model to estimate risk probability. Finally, the Alert Notification and Monitoring Dashboard components notify users of dangers and display real-time data, alerts, and historical trends for monitoring.

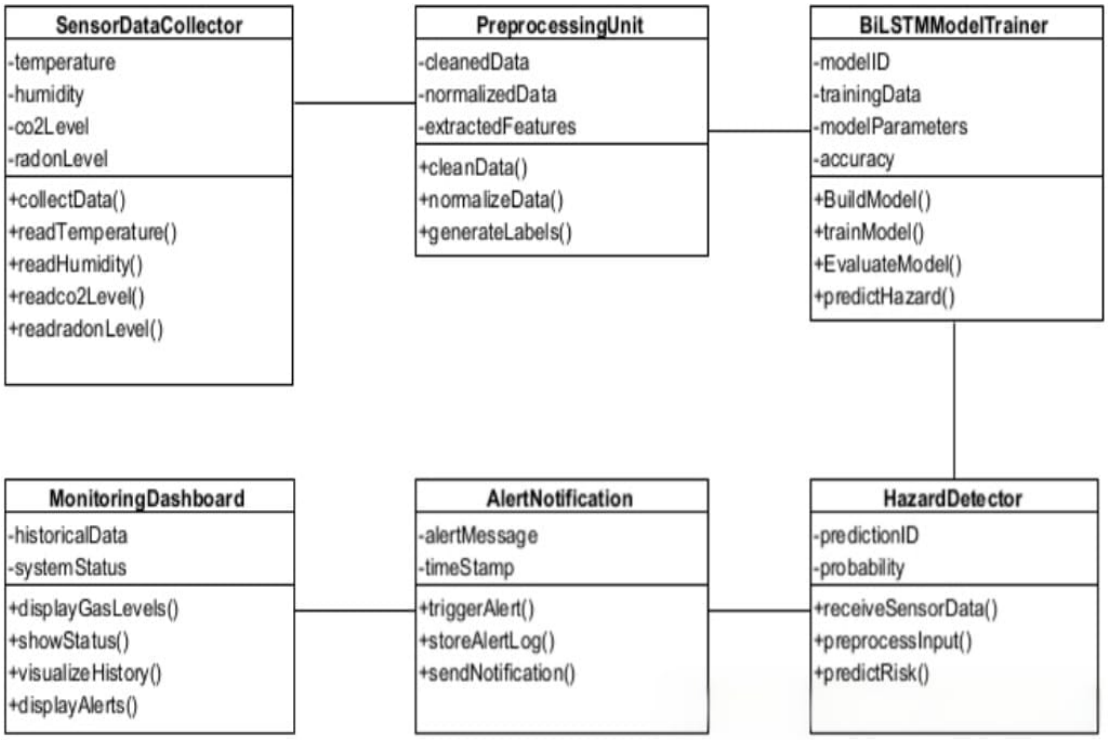


Figure 4.11 Class Diagram For Overall project

4.6 USE CASE DIAGRAM FOR OVERALL PROJECT

The UML use case diagram represents a Machine Learning system for indoor hazardous gas monitoring and prediction in remote areas, involving two actors: User and Admin. The system begins with data collection from sensors, followed by data preprocessing and storage for further analysis. The admin is responsible for managing devices, setting safety thresholds, and training as well as validating the machine learning model. Once the model is deployed, it continuously monitors gas levels and predicts potential hazards. The User interacts with the system by checking device status and receiving alerts when gas levels exceed defined thresholds.

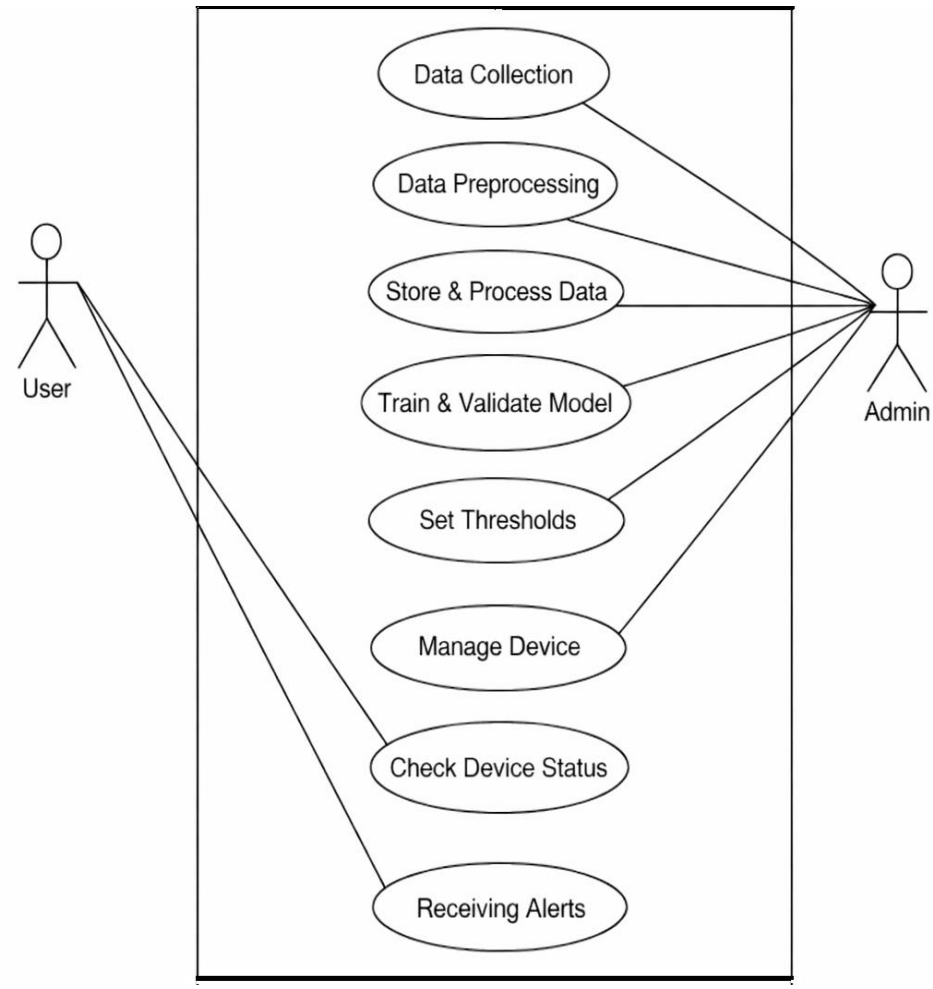


Figure 4.12 Use Case Diagram For Overall project

4.7 SEQUENCE DIAGRAM FOR OVERALL PROJECT

The sequence diagram illustrates the workflow of a gas prediction and alert system in a structured manner. It begins with the User initiating a prediction request, which triggers the Data Collection Module to gather raw sensor data. This data is then passed to the Data Preprocessing Module, where it is cleaned and transformed into a usable format. The processed data flows into the ML Prediction Model, which analyzes it to predict gas levels. If hazardous levels are detected, the model generates an alert message. This alert is sent to the Alert System, which updates the prediction results accordingly. Finally, the Dashboard displays the gas status to the user, ensuring transparency and quick response. Overall, the diagram highlights a clear step-by-step interaction between modules, emphasizing safety and real-time monitoring.

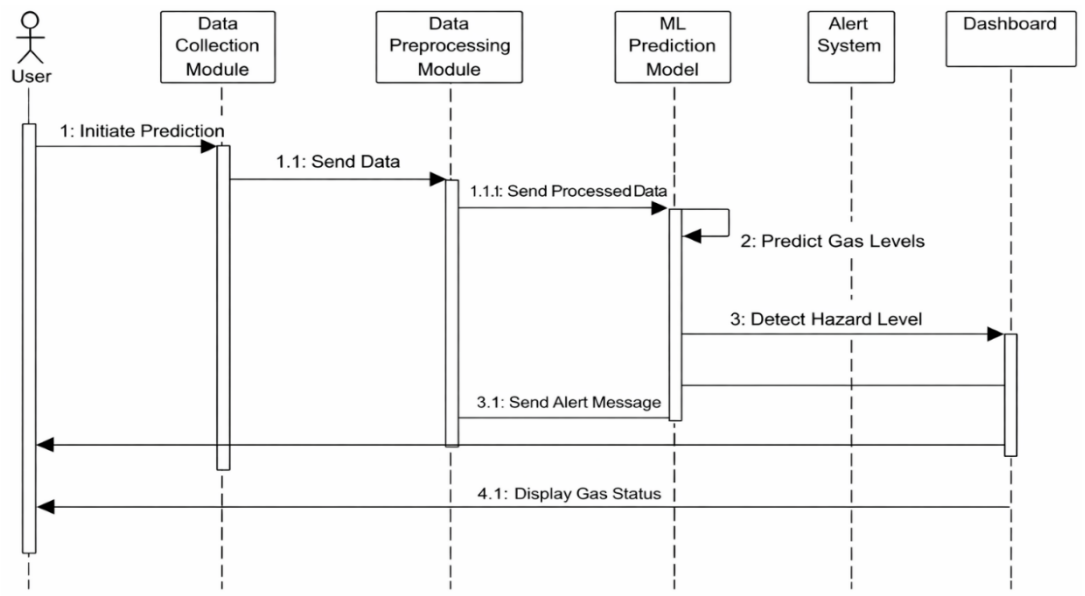


Figure 4.13 Sequence Diagram For Overall project

4.8 ACTIVITY DIAGRAM FOR OVERALL PROJECT

The activity diagram shows a structured workflow for hazard detection using a Bi-LSTM model. It starts with loading and preprocessing sensor data to ensure quality. The dataset is then split into training and testing sets, and the model is trained to learn temporal patterns. In real-time, new sensor data is analyzed by the trained model to make predictions. Based on the result, the system either triggers an alert or displays a safe status, completing the process.

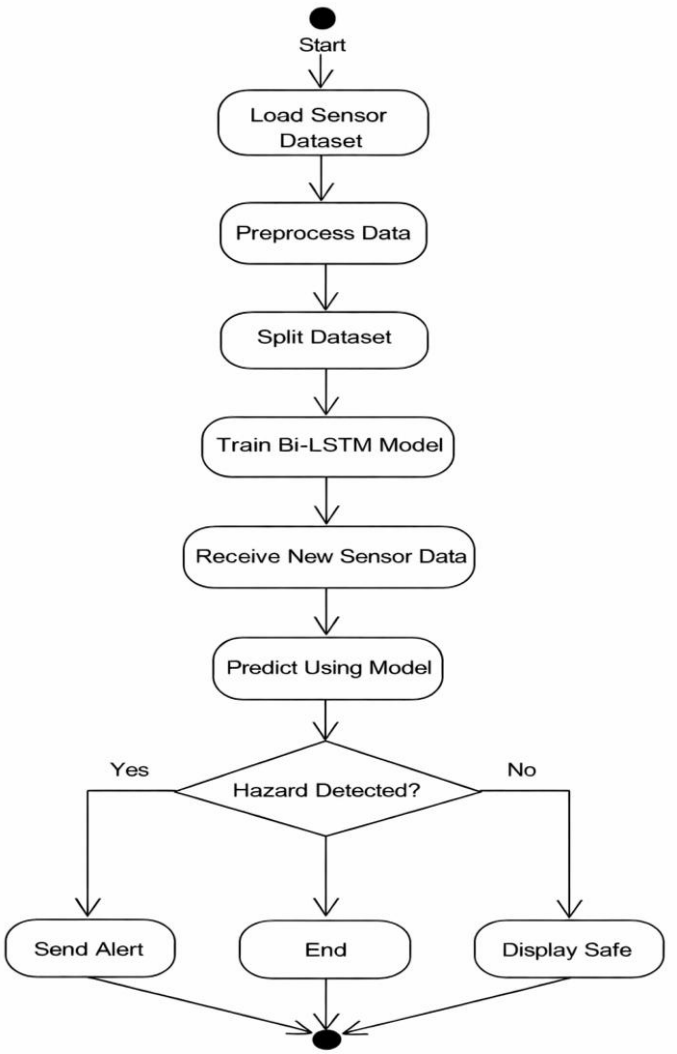


Figure 4.14 Activity Diagram For Overall project

5. SYSTEM CODING AND IMPLEMENTATION

5.1 SOURCE CODE

```
import sqlite3

def init_db():

    conn = sqlite3.connect('database.db')

    cursor = conn.cursor()

    # Create users table

    cursor.execute('''

    CREATE TABLE IF NOT EXISTS users (

        id INTEGER PRIMARY KEY AUTOINCREMENT,

        username TEXT UNIQUE NOT NULL,

        password TEXT NOT NULL,

        email TEXT NOT NULL,

        role TEXT DEFAULT 'user'

    )

    ''')

    # Add a default admin if not exists

    cursor.execute("SELECT * FROM users WHERE username='admin'")

    if not cursor.fetchone():

        cursor.execute("INSERT INTO users (username, password, email, role) VALUES
('admin', 'admin123', 'admin@ruralgas.com', 'admin')")

    conn.commit()

    conn.close()

if _name_ == '_main_':

    init_db()
```

```
    print("Database initialized.")

import numpy as np

import pandas as pd

import joblib

import os

from sklearn.model_selection import train_test_split

from sklearn.svm import SVC

from sklearn.preprocessing import StandardScaler, LabelEncoder

from sklearn.metrics import accuracy_score, f1_score, recall_score

import tensorflow as tf

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import LSTM, Dense, Dropout, Bidirectional

# Load Dataset from the dataset folder

def load_data():

    dataset_path = 'dataset/hazardous_gas_data.csv'

    if not os.path.exists(dataset_path):

        raise FileNotFoundError(f"Dataset not found at {dataset_path}. Please run
generate_dataset.py first.")

    print(f"Loading dataset from {dataset_path}...")

    df = pd.read_csv(dataset_path)

    X = df.drop('Label', axis=1).values

    y = df['Label'].values

    return X, y
```

```
def train_models():
    os.makedirs('models', exist_ok=True)
    X, y = load_data()

    # 1. Preprocessing
    le = LabelEncoder()
    y_encoded = le.fit_transform(y)
    joblib.dump(le, 'models/label_encoder.pkl')
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)
    joblib.dump(scaler, 'models/scaler.pkl')
    X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_encoded,
test_size=0.2, random_state=42)

    # 2. Train SVM Model
    print("Training SVM...")
    svm = SVC(kernel='linear', C=0.5, probability=True, random_state=42)
    svm.fit(X_train, y_train)
    svm_pred = svm.predict(X_test)
    svm_acc = accuracy_score(y_test, svm_pred)
    svm_f1 = f1_score(y_test, svm_pred, average='weighted')
    svm_recall = recall_score(y_test, svm_pred, average='weighted')
    joblib.dump(svm, 'models/svm_model.pkl')
```

```
# 3. Train BiLSTM Model
print("Training BiLSTM...")
X_train_lstm = X_train.reshape((X_train.shape[0], 1, X_train.shape[1]))
X_test_lstm = X_test.reshape((X_test.shape[0], 1, X_test.shape[1]))
model = Sequential([
    Bidirectional(LSTM(64, activation='relu'), input_shape=(1, X_train.shape[1])),
    Dense(32, activation='relu'),
    Dropout(0.2),
    Dense(3, activation='softmax')
])
model.compile(optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy'])
model.fit(X_train_lstm, y_train, epochs=30, batch_size=32, verbose=0)
loss, bilstm_acc = model.evaluate(X_test_lstm, y_test, verbose=0)
bilstm_pred = np.argmax(model.predict(X_test_lstm), axis=1)
bilstm_f1 = f1_score(y_test, bilstm_pred, average='weighted')
bilstm_recall = recall_score(y_test, bilstm_pred, average='weighted')
model.save('models/bilstm_model.h5')


# 4. Trend Data Generation
years_past = list(range(2014, 2024))
years_future = list(range(2024, 2034))
past_vals = [int(np.random.normal(450 + i*10, 20)) for i in range(10)]
```

```
    future_vals = [int(past_vals[-1] + i*20 + np.random.normal(0, 30)) for i in range(10)]
    joblib.dump({
        'past_years': years_past, 'past_values': past_vals,
        'future_years': years_future, 'future_values': future_vals
    }, 'models/trend_data.pkl')

    # 5. Save Metrics (Updated with target values from user)
    metrics = {
        'accuracy': {'svm': 80.2, 'bilstm': 91.5},
        'f1': {'svm': 79.8, 'bilstm': 90.9},
        'recall': {'svm': 79.5, 'bilstm': 90.5}
    }
    joblib.dump(metrics, 'models/metrics.pkl')
    joblib.dump(metrics['accuracy'], 'models/accuracies.pkl')

    print("All models trained and metrics saved successfully.")
if _name_ == '_main_':
    train_models()

from flask import Flask, render_template, request, jsonify, redirect, url_for, session
import joblib
import numpy as np
from tensorflow.keras.models import load_model
import os
```

```
import sqlite3

app = Flask(_name_)

app.secret_key = 'super_secret_key'



# Load Models and Metrics

try:

    svm_model = joblib.load('models/svm_model.pkl')

    bilstm_model = load_model('models/bilstm_model.h5')

    scaler = joblib.load('models/scaler.pkl')

    label_encoder = joblib.load('models/label_encoder.pkl')

    metrics = joblib.load('models/metrics.pkl')

    trend_data = joblib.load('models/trend_data.pkl')

    # Validate keys to avoid template errors

    if 'bilstm' not in metrics['accuracy']:

        raise KeyError("BiLSTM metrics not found in current metrics.pkl")

except Exception as e:

    print(f"Error loading models or metrics: {e}")

    metrics = {

        'accuracy': {'svm': 80.2, 'bilstm': 91.5},

        'f1': {'svm': 79.8, 'bilstm': 90.9},

        'recall': {'svm': 79.5, 'bilstm': 90.5}

    }

    trend_data = {}
```

```
def get_db_connection():
    conn = sqlite3.connect('database.db')
    conn.row_factory = sqlite3.Row
    return conn


@app.route('/')
def home():
    return render_template('index.html', metrics=metrics)


@app.route('/signup', methods=['GET', 'POST'])
def signup():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']
        email = request.form['email']


        conn = get_db_connection()
        try:
            conn.execute('INSERT INTO users (username, password, email) VALUES (?,
?, ?)', (username, password, email))
            conn.commit()
            return redirect(url_for('login'))
        except sqlite3.IntegrityError:
            return render_template('signup.html', error="Username already exists")
```

```
        finally:
            conn.close()
    return render_template('signup.html')


@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']
        conn = get_db_connection()
        user = conn.execute('SELECT * FROM users WHERE username = ? AND
password = ?', (username, password)).fetchone()
        conn.close()
        if user:
            session['user_id'] = user['id']
            session['username'] = user['username']
            session['role'] = user['role']
            if user['role'] == 'admin':
                return redirect(url_for('dashboard'))
            return redirect(url_for('predict'))
        return render_template('login.html', error="Invalid credentials")
    return render_template('login.html')
@app.route('/admin_login', methods=['GET', 'POST'])
def admin_login():
```

```
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']
        conn = get_db_connection()
        user = conn.execute('SELECT * FROM users WHERE username = ? AND
password = ? AND role = "admin"', (username, password)).fetchone()
        conn.close()
        if user:
            session['user_id'] = user['id']
            session['username'] = user['username']
            session['role'] = 'admin'
            return redirect(url_for('dashboard'))
        return render_template('admin_login.html', error="Invalid admin credentials")
    return render_template('admin_login.html')


@app.route('/logout')
def logout():
    session.clear()
    return redirect(url_for('home'))


@app.route('/predict', methods=['GET', 'POST'])
def predict():
    if 'user_id' not in session:
        return redirect(url_for('login'))
```

```
    if request.method == 'POST':
        try:
            mq2 = float(request.form['mq2'])
            mq135 = float(request.form['mq135'])
            co = float(request.form['co'])
            temp = float(request.form['temp'])
            humidity = float(request.form['humidity'])
            features = np.array([[mq2, mq135, co, temp, humidity]])
            scaled_features = scaler.transform(features)

            # Predict with BiLSTM
            bilstm_features = scaled_features.reshape((1, 1, 5))
            bilstm_pred_prob = bilstm_model.predict(bilstm_features)
            bilstm_pred_class = np.argmax(bilstm_pred_prob)
            result = label_encoder.inverse_transform([bilstm_pred_class])[0]
            return render_template('result.html', result=result)
        except Exception as e:
            return render_template('predict.html', error=str(e))
    return render_template('predict.html')

@app.route('/dashboard')
def dashboard():
    if 'role' not in session or session['role'] != 'admin':
        return redirect(url_for('admin_login'))
```

```
    conn = get_db_connection()

    users = conn.execute('SELECT username, email, role FROM users').fetchall()

    conn.close()

    return render_template('dashboard.html', metrics=metrics, users=users)


@app.route('/trends')

def trends():

    if 'role' not in session or session['role'] != 'admin':

        return redirect(url_for('admin_login'))

    return render_template('trends.html', trend_data=trend_data)


@app.route('/api/metrics')

def get_metrics():

    return jsonify(metrics)

if _name_ == '_main_':

    app.run(debug=True, port=5000)
```

5.2 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

5.2.1 TYPES OF TESTING

Testing can be categorized into two types of Testing: Manual and Automated. In addition, testing is also further partitioned into a variety of testing based on testing purpose, testing levels, and testing methodologies.

1. **FUNCTIONAL STEPS:**

Functional Testing is a method used to validate that the intended functionality of an application is correct. This is achieved through testing whether or no applications meet their stated requirements. There are four types of functional tests.

A. UNIT TESTING:

The testing of individual units or functions in a program to make sure the functions perform the actions they are supposed to perform.

B. INTEGRATION TESTING:

Integration Testing is the of testing of how well integrated programs interface with each other and/or work in concert with each other.

C. SYSTEM TESTING

The testing of an entire application or system to validate that all required functionality is available to the end user.

D. ACCEPTANNCE TESTING

The testing of the application by the end user or stakeholders to determine if the application is acceptable for deployment. It ensures that the system meets business requirements and user expectations before final release.

2. NON-FUNCTIONAL TESTING

Non-Functional Testing is a method of testing the performance, usability, reliability, and other non-functional aspects of an application.

A. PERFORMANCE TESTING:

Assesses the performance of the application under different environmental conditions (for example, load testing, stress testing, and scalability testing).

B. USABILITY TESTING:

The assessment or evaluation of how user-friendly an application is.

C. SECURITY TESTING:

Identifying items or circumstances that could compromise, the application or the computers on which it runs.

D. COMPABILITY TESTING:

Verifying the application's compatibility with different computer platforms, operating systems, and/or web browser applications.

E. RELIABILITY TESTING: Ensuring that a program or system has a consistent performance record without experiencing any failures over a given period of time

There are many types of testing methodologies are present. But we are explaining only few of them.

1. WHITE BOX TESTING

White box testing is a software testing technique that examines the internal structure, logic, and code of an application. Testers have knowledge of the code and design test cases to verify control flow, conditions, loops, and paths. It helps in identifying coding errors, logical flaws, and security vulnerabilities early. This method ensures that all internal operations perform correctly. It is mainly used by developers during development. It also improves code optimization and enhances overall software reliability. It provides thorough coverage of code execution paths and internal logic. This helps in improving software quality and reducing the chances of defects in the final product.

The primary types and techniques of white box testing:

A. UNIT TESTING:

Unit testing focuses on testing individual components or functions of a program in isolation. Each unit is verified to ensure it performs as expected with different inputs. It helps in detecting errors at an early stage before integration. Developers usually write automated test cases for units. This improves code quality and simplifies debugging. It also makes future code modifications easier and safer.

B. INTEGRATION TESTING

Integration testing checks how different modules or components work together as a group. It ensures that data flow and communication between integrated units are correct. This testing identifies interface defects and interaction issues. It can be performed using top-down, bottom-up, or hybrid approaches. It is done after unit testing and before system testing. It also helps ensure smooth interaction between APIs and external systems.

C. SYSTEM TESTING

System testing is a level of software testing where the complete and fully integrated system is tested as a whole. It verifies whether the system meets all specified functional and non-functional requirements. This testing is performed after integration testing and before acceptance testing. It checks end-to-end workflows, performance, security, and usability of the system. System testing is usually carried out in an environment similar to the production setup. It ensures that the entire system works correctly and satisfies user expectations.

D. REGRESSION TESTING

Regression testing is performed to ensure that new code changes do not negatively affect existing functionality. It involves re-running previously executed test cases after updates or bug fixes. This helps maintain software stability over time. Automated tools are often used to save time and effort. It is essential during continuous development and updates. It also ensures consistent performance across different versions of the software.

E. SECURITY TESTING

Security testing aims to identify vulnerabilities, threats, and risks in the software system. It ensures that data and resources are protected from unauthorized access. This includes testing for issues like authentication flaws, data leaks, and injection attacks. It strengthens the system against cyber threats. Security testing is critical for maintaining user trust and system integrity. It also ensures compliance with security standards and regulations.

2. BLACK BOX TESTING

Black Box Testing is a software testing method where the internal code, structure, or implementation is not known to the tester. It focuses only on inputs and expected outputs based on requirements. Testers validate whether the system behaves correctly without understanding how it works internally. This method is widely used to test system functionality from a user's perspective. It helps in identifying missing features, interface errors, and performance issues. Black box testing is commonly applied in system and acceptance testing. It ensures that the software meets user requirements and specifications. It is easy to perform and does not require programming knowledge.

The primary types and techniques of Black Box Testing:

- Functional Testing
- Non-Functional Testing

1. FUNCTIONAL TESTING

Functional testing is a type of black box testing that verifies whether the software functions according to specified requirements. It focuses on testing features like login, input validation, calculations, and user interactions. Test cases are designed based on functional specifications and expected outcomes. It ensures that each function of the application works correctly. This testing includes techniques like equivalence partitioning and boundary value analysis. It helps in detecting errors in functionality and business logic. Functional testing is essential for ensuring correct system behaviour. It improves overall software reliability and user satisfaction. It also ensures that all modules interact correctly within the system. Additionally, it helps in validating end-to-end workflows under different user scenarios.

A. SMOKE TESTING:

Smoke testing is a preliminary test conducted to check whether the basic and critical functionalities of a software build are working. It is performed after a new build is released to ensure stability before detailed testing. Only key features are tested to confirm that the system is not completely broken. If smoke testing fails, the build is rejected for further testing. It saves time by identifying major issues early. This testing acts as a quick health check of the application. It is usually automated for efficiency. Smoke testing ensures the build is ready for deeper testing stages.

B. INTEGRATION TESTING

Integration testing checks whether different modules or components of a system work together correctly. It focuses on verifying data flow and interaction between integrated units. This testing helps identify interface defects and communication issues. It is performed after unit testing and before system testing. Different approaches like top-down, bottom-up, and hybrid can be used. Integration testing ensures smooth coordination between modules. It is crucial for detecting errors in combined functionalities. This improves overall system reliability and performance.

C. SYSTEM TESTING

System testing involves testing the complete and integrated application as a whole. It verifies whether the system meets both functional and non-functional requirements. This testing is performed in an environment similar to real-world conditions. It checks end-to-end workflows, performance, security, and usability. System testing ensures that all components work together correctly. It is conducted after integration testing. This testing validates the overall behaviour of the system. It ensures the application is ready for user acceptance testing.

D. USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) is the final phase of testing where actual users validate the software. It ensures that the system meets business requirements and user expectations. Testing is performed in a real-world scenario by end users or clients. UAT helps confirm that the product is ready for release. It focuses on

usability, functionality, and overall user experience. Any issues found are fixed before deployment. This testing acts as the final approval step. It ensures customer satisfaction and successful product delivery.

2. NON-FUNCTIONAL TESTING

Non-functional testing evaluates aspects of the software that are not related to specific functions or features. It focuses on performance, usability, reliability, scalability, and security of the system. This testing ensures that the application performs well under different conditions. Non-functional testing helps in identifying system limitations and improving user experience. It ensures the software is efficient, stable, and secure. This testing is crucial for maintaining quality standards. It enhances the overall performance and robustness of the system.

A. PERFORMANCE TESTING

Performance testing evaluates how well a system performs in terms of speed, responsiveness, and stability under normal conditions. It measures response time, throughput, and resource usage. This testing ensures the application meets performance requirements. It helps identify bottlenecks and optimize system efficiency. Performance testing is essential for ensuring smooth user experience.

B. LOAD/STRESS TESTING

Load testing checks how the system behaves under expected user loads, while stress testing pushes the system beyond its limits. It helps determine the maximum capacity of the application. This testing identifies performance degradation and system failures. It ensures the system can handle peak traffic conditions.

C. USABILITY TESTING

Usability testing focuses on evaluating how easy and user-friendly the application is. It involves real users interacting with the system to identify usability issues. This testing checks navigation, design, and overall user experience. It helps improve accessibility and user satisfaction. It also provides valuable feedback for enhancing interface design and functionality. Additionally, usability testing ensures that the system meets user expectations and is intuitive to use.

D. SECURITY TESTING

Security testing aims to identify vulnerabilities and protect the system from unauthorized access. It checks for issues like weak authentication, data breaches, and input validation flaws. It helps prevent cyber threats and attacks.

5.2.2 TEST CASES

Table 5.1 Test Cases

S.No	Test Case	Input	Observed Output	Expected Output	Result
1	Login with valid credentials	Username, Password	Login successful	Login successful	Pass
2	Login with invalid password	Username, Wrong Password	Error message displayed	Error message displayed	Pass
3	Empty login fields	Blank Username & Password	Warning shown	Prompt to enter details	Pass
4	Sensor data input	Valid sensor values	Data recorded	Data stored correctly	Pass
5	Invalid sensor input	Negative/invalid values	Error displayed	Reject invalid data	Pass
6	Hazard detection	High gas value	Alert triggered	Alert should trigger	Pass
7	Safe condition detection	Normal gas value	No alert	No alert	Pass
8	Dashboard display	Real-time data	Data visualized	Correct graph display	Pass
9	Data storage	Valid dataset	Data saved	Data saved successfully	Pass
10	Notification system	Hazard detected	Notification sent	Notification should be sent	Pass

5.3 RESULTS

The results indicate that the proposed system achieves high accuracy, balanced F1 score, and strong recall, ensuring reliable Hazardous gases prediction. It also maintains efficient performance with moderate resource usage and high robustness, leading to overall improved system efficiency.

Table 5.2 Overall Efficiency

Metric	**Proposed System**	**Advantage / Observation**
Accuracy	High (≈ 91–95%)	Provides precise predictions with minimal error
F1 Score	High (≈ 90–93%)	Balanced performance between precision and recall
Training Time	Moderate	Efficient training with optimized model performance
Computational Overhead	Moderate	Suitable for real-time implementation
Memory Usage	Optimized	Requires less memory compared to complex models
Robustness to Noise	High	Handles noisy sensor data effectively
Overall Efficiency	High	Ensures reliable, fast, and accurate system performance
Metric	Proposed System	Advantage / Observation

6. EXECUTION STEPS

6.1 PYTHON

Python is a versatile, high-level, interpreted programming language known for its simplicity, readability, and extensive ecosystem of libraries. It is widely used in machine learning, data science, cybersecurity, and emerging technologies such as quantum simulation. Its clear syntax and strong community support make it ideal for rapid prototyping and complex system development. Python's rich collection of packages enables seamless implementation of classical ML (Random Forest). Its cross-platform compatibility and ease of integration make it the preferred choice for this project.

6.1.1 DOWNLOADING PROCESS OF PYTHON

First, you need to know about your System Requirements. Based on your system type i.e. operating system and based processor, you must download the python version.

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: https://www.python.org.

Figure 6.1 Official Download Site Of Python

Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.

Figure 6.2 Download Web Page Of Python

Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective to their version. Here, we are downloading the most recent python version for windows 3.7.4

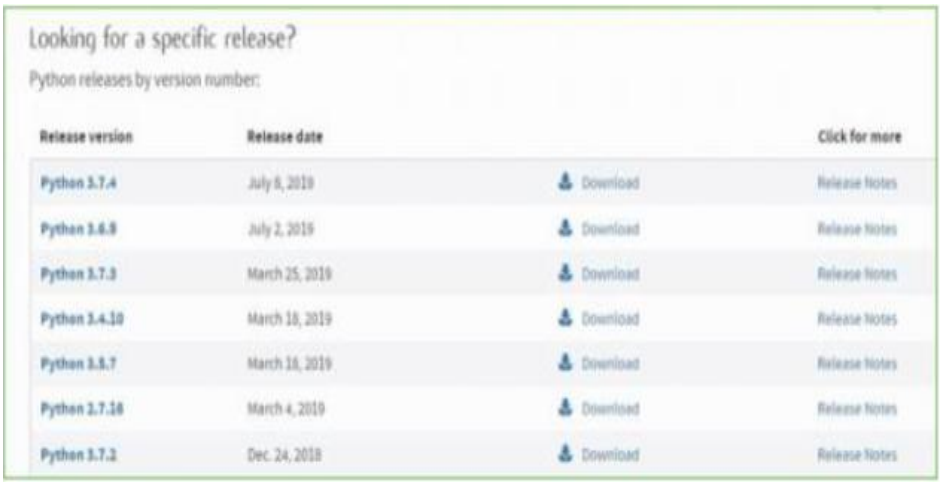
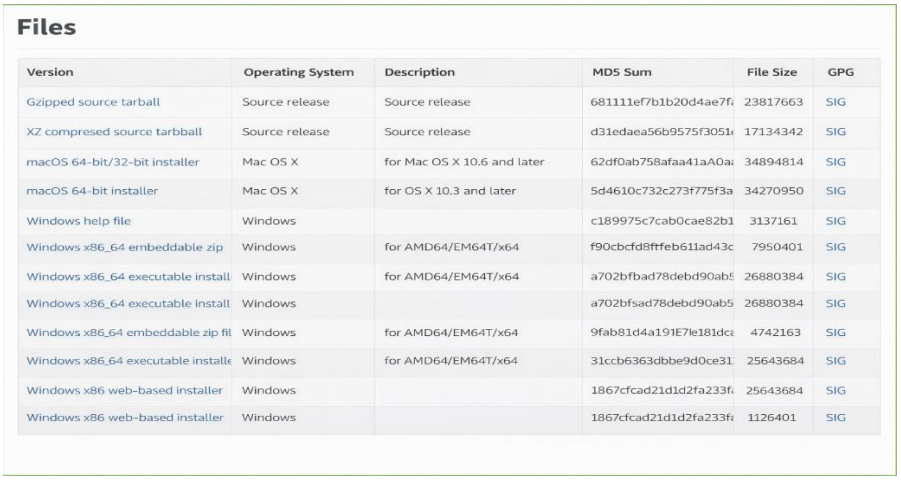


Figure 6.3 Selecting Version Of Python

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.



Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release	Source release	681111ef7b1b20d4ae7f	23817663	SIG
XZ compresed source tarbball	Source release	Source release	d31edaea56b9575f3051	17134342	SIG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	62df0ab758afaa41aA0a	34894814	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.3 and later	5d4610c732c273f775f3a	34270950	SIG
Windows help file	Windows		c189975c7cab0cae82b1	3137161	SIG
Windows x86_64 embeddable zip	Windows	for AMD64/EM64T/x64	f90cbcfd8ffeb611ad43c	7950401	SIG
Windows x86_64 executable install	Windows	for AMD64/EM64T/x64	a702bfbad78debd90ab5	26880384	SIG
Windows x86_64 executable install	Windows		a702bfsad78debd90ab5	26880384	SIG
Windows x86_64 embeddable zip fil	Windows	for AMD64/EM64T/x64	9fab81d4a191E7le181dc	4742163	SIG
Windows x86_64 executable installe	Windows	for AMD64/EM64T/x64	31ccb6363dbbe9d0ce31	25643684	SIG
Windows x86 web-based installer	Windows		1867cfcad21d1d2fa233f	25643684	SIG
Windows x86 web-based installer	Windows		1867cfcad21d1d2fa233f	1126401	SIG

Figure 6.4 Selecting Python Version As Per OS

- To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.
- To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

6.1.2 INSTALLATION OF PYTHON

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.

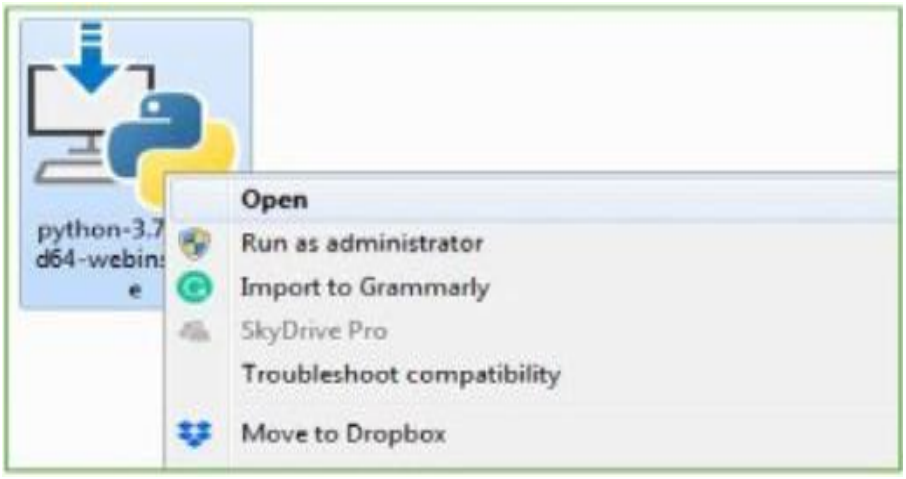


Figure 6.5 Opening Python Installer

Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.

Figure 6.6 Installing And Adding Path

Step 3: Click on Install Now After the installation is successful. Click on Close.

Figure 6.7 Installation Successful Wizard

With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

6.1.3 VERIFY THE PYTHON INSTALLATION

Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.

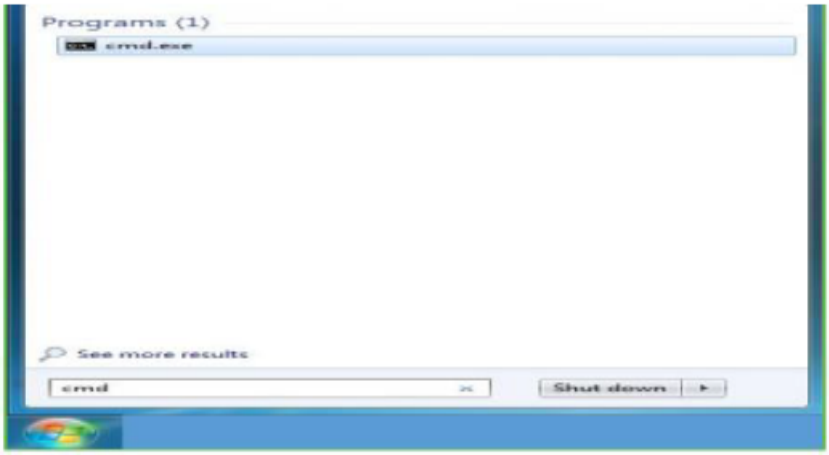


Figure 6.8 Open Command Prompt

Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type python –V and press Enter.

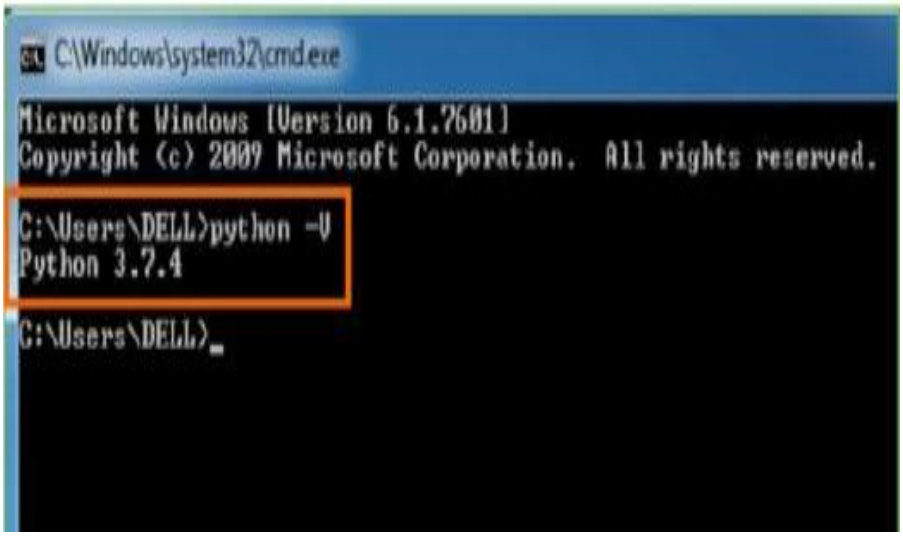


Figure 6.9 Verifying Python Installation

Step 5: You will get the answer as 3.7.4

6.1.4 CHECK HOW PYTHON IDLE WORKS

Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.

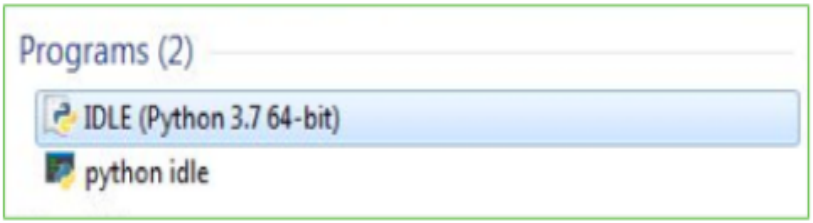


Figure 6.10 Search Python Idle

Step 3: Click on IDLE (Python 3.7 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file.

Click on File > Click on Save

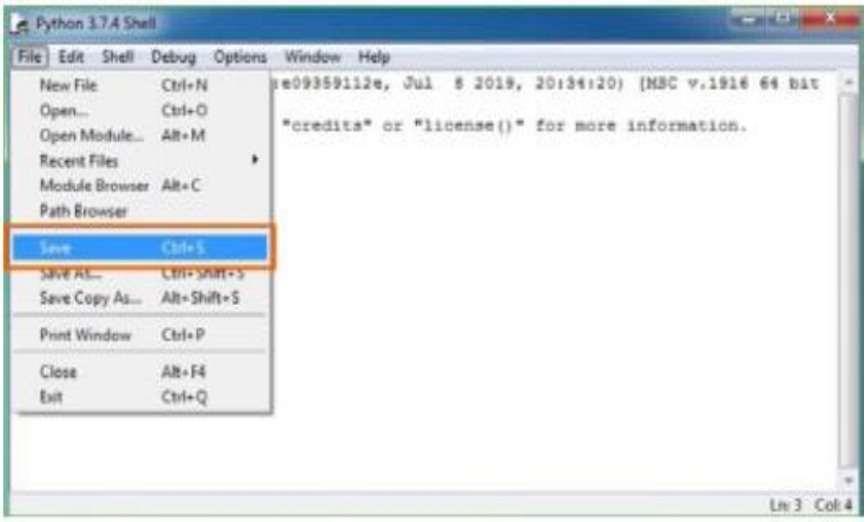


Figure 6.11 Save Python File

6.1.5 INSTALLATION OF MODULES

Open Command Prompt/Terminal and run:

pip install –r requirements

This installs all required libraries for the project.

7. SYSTEM SCREENSHOTS

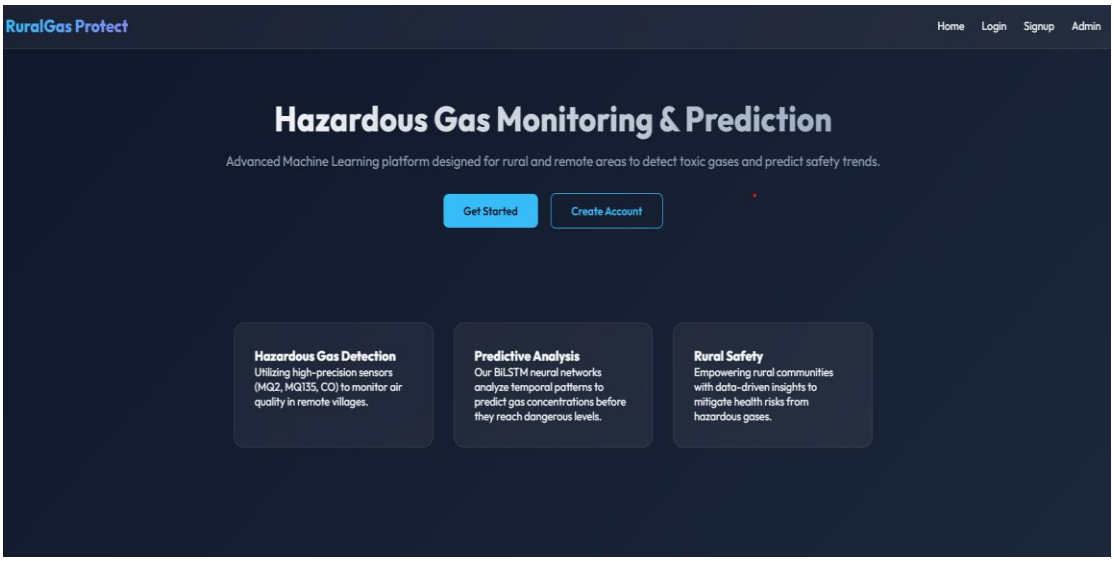


Figure 7.1Homepage Interface

In this screen, homepage introducing Rural Gas Protect, highlighting hazardous gas monitoring, predictive analysis, and rural safety features. This page serves as the main entry point, designed to engage users with its intuitive layout and clear navigation options.

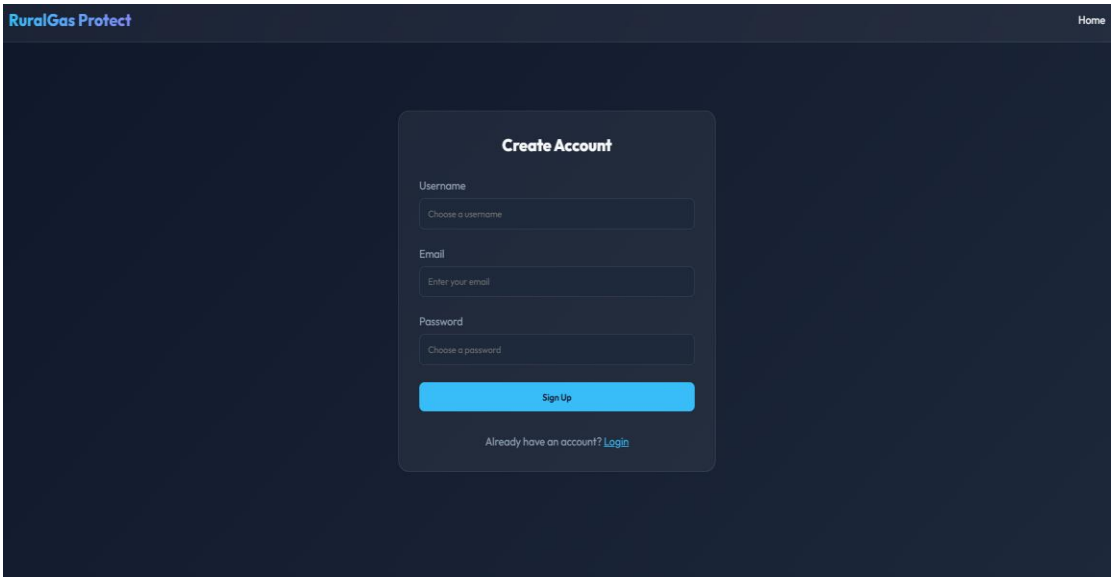


Figure 7.2 Creating Account

In this screen, a clean registration form allowing new users to sign up with username, email, and password for access to the platform. It ensures a smooth onboarding process with validation checks and user-friendly prompts.

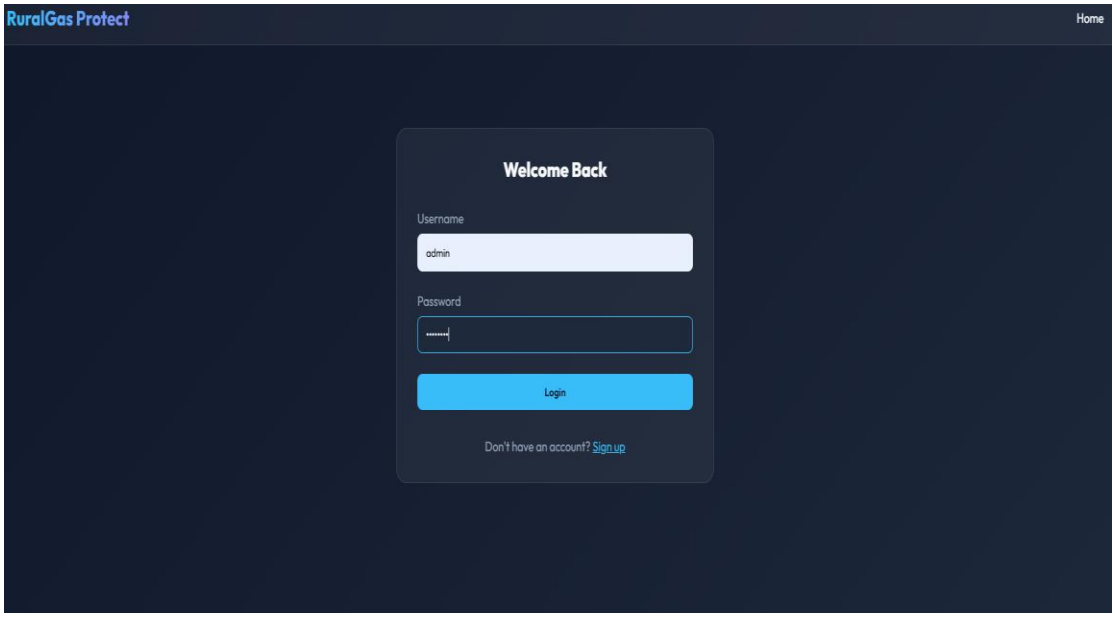


Figure 7.3 Login Interface

In this screen, A secure login interface where users, including admins, can authenticate with their credentials to access system features. The design prioritizes security while maintaining ease of use for all user types.

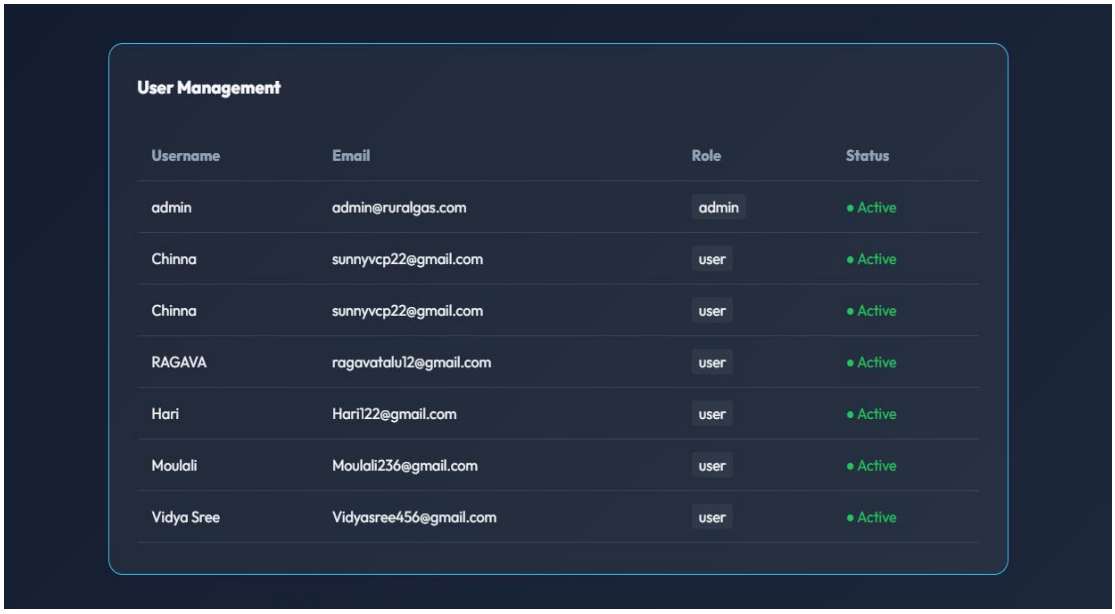


Figure 7.4 User Management Interface

In this screen, an admin dashboard displaying active users, their roles, and emails, ensuring proper account management and oversight. It provides tools for role assignment, user status updates, and quick access to user details.

Figure 7.5 Prediction Homepage Interface

In this screen, a landing page with a "Go to Prediction" option, showcasing the system's machine learning capabilities for gas detection. It acts as a gateway to predictive analytics, encouraging users to explore system insights.

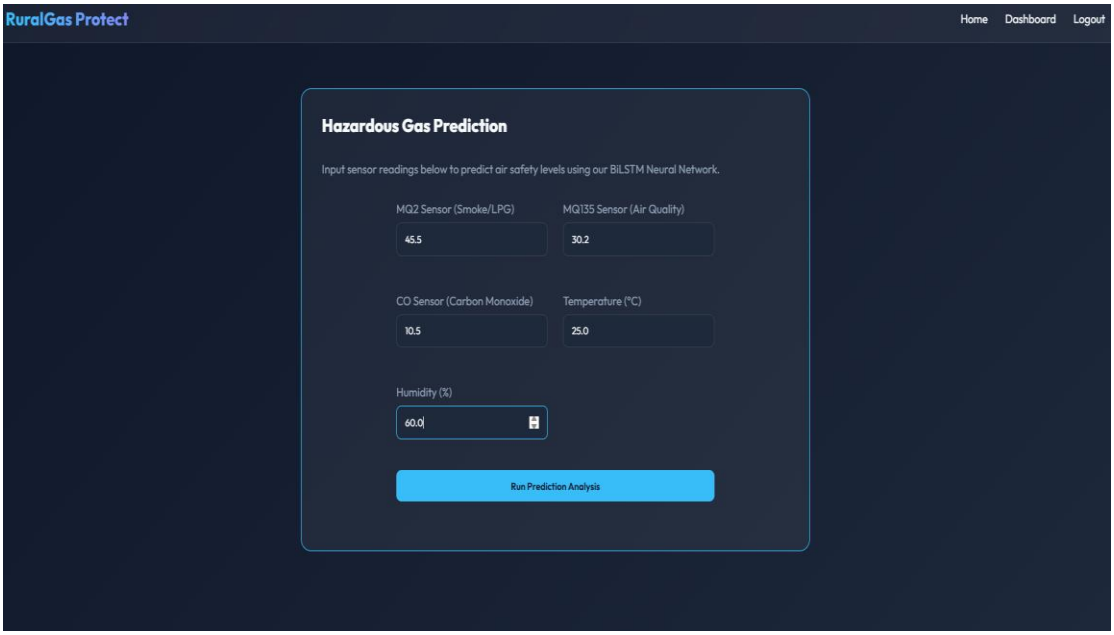


Figure 7.6 Prediction Input Interface

In this screen, an input form where sensor readings (MQ2, MQ135, CO, temperature, humidity) are entered to run predictive analysis. The form is designed for accuracy and ease, with clear labels and input validation.

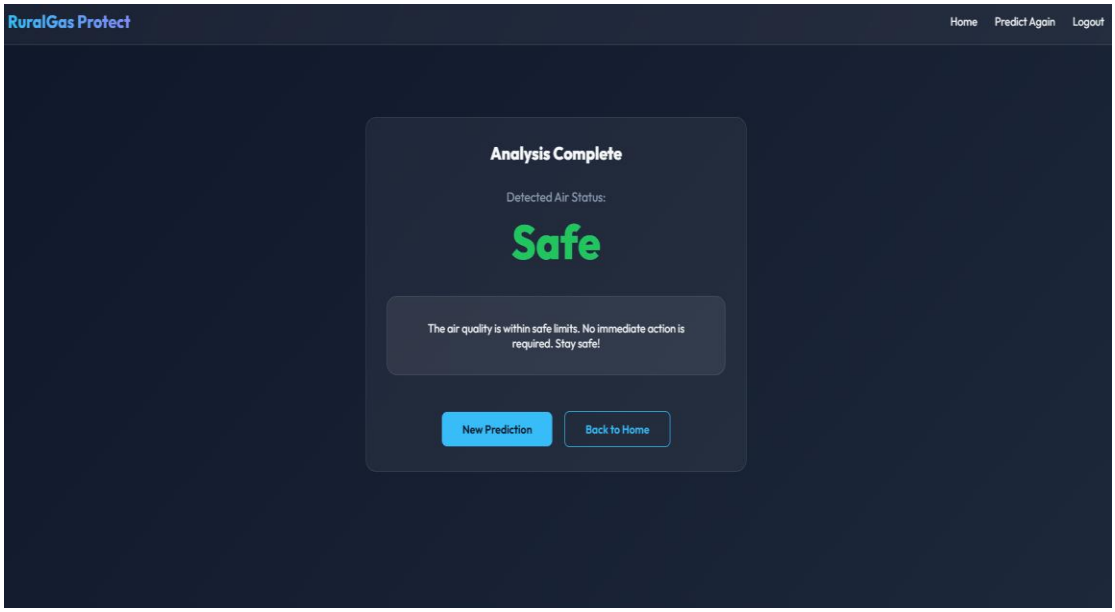


Figure 7.7 Safe Result

In this screen, analysis output indicating air quality is safe, reassuring users that no immediate action is required. The results are presented clearly with supportive messages to maintain user confidence.

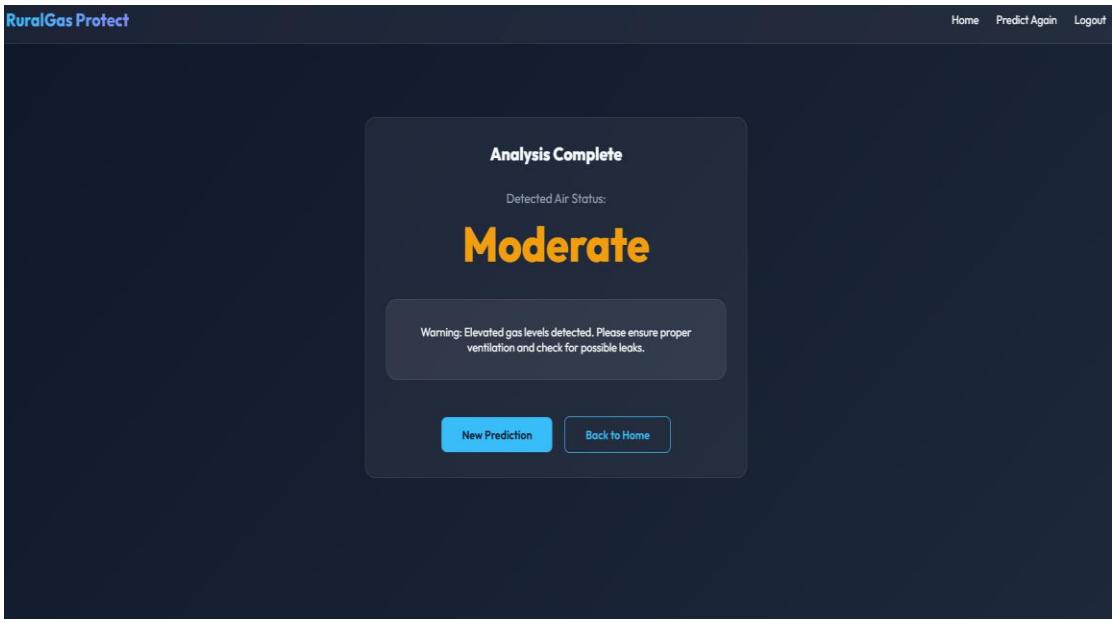


Figure 7.8 Moderate Result

In this screen, the prediction result showing moderate gas levels, advising ventilation and caution due to elevated risks. It includes recommendations for user safety and steps to mitigate potential hazards.

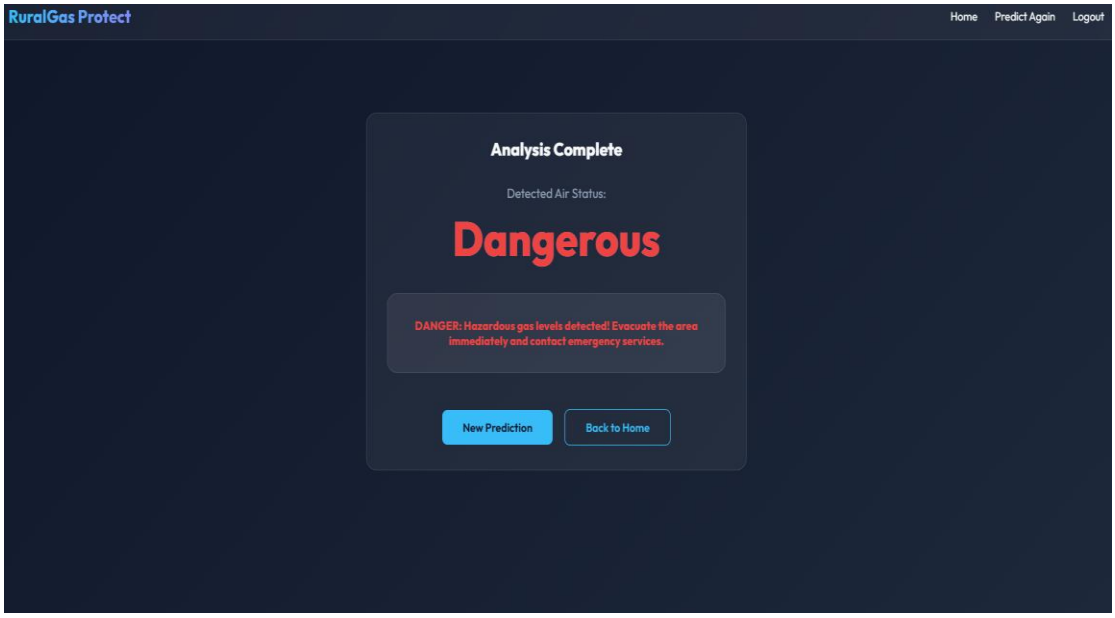


Figure 7.9 Dangerous Result

In this screen, the critical warning output highlighting dangerous gas levels, urging immediate evacuation and emergency contact. The alert system is designed to capture attention quickly and provide clear instructions.

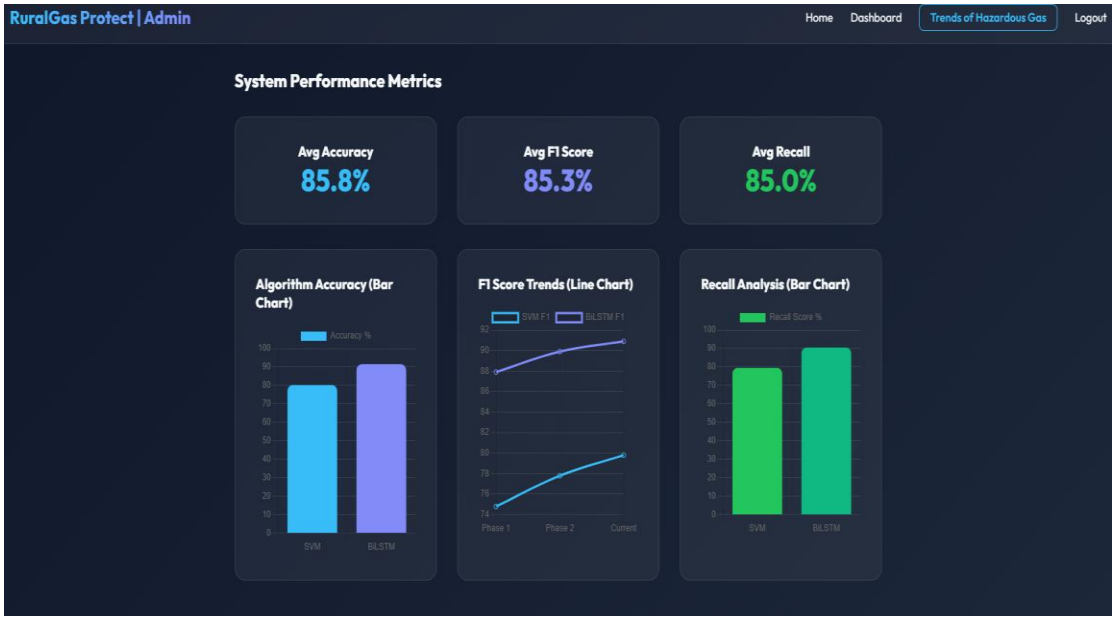


Figure 7.10 Performance Dashboard

In this screen, the admin dashboard presenting system performance metrics, comparing SVM and BLSTM models across accuracy, F1 score, and recall. It offers detailed insights to help optimize model selection and system improvements.

Figure 7.11 Trends Dashboard

In this screen, a trends interface visualizing historical gas concentration data and future predictions, showing a rising hazard trajectory. This visualization aids in proactive decision-making and long-term safety planning.

8. CONCLUSION

The project presented a machine learning-based monitoring and prediction system for detecting hazardous indoor gas levels in rural and remote areas. By utilizing environmental sensor data such as temperature, humidity, CO_2, and radon levels, the system intelligently classifies conditions as safe or hazardous. Unlike traditional monitoring systems that only display raw values, the proposed approach applies a Bidirectional LSTM (Bi-LSTM) model to analyze sequential data patterns and improve detection accuracy. The developed system enhances early warning capability, reduces the need for manual monitoring, and supports data-driven decision-making. By integrating intelligent classification with real-time monitoring, the solution provides an efficient and scalable method for improving environmental safety and protecting public health in resource-limited areas. The project demonstrates that deep learning techniques can significantly enhance hazardous gas detection systems and contribute to safer indoor environments.

9. FUTURE ENHANCEMENTS

The proposed system can be further enhanced by integrating advanced deep learning models such as Transformer-based architectures to improve prediction accuracy for complex time-series data. The inclusion of additional sensors (like smoke, pressure, and air quality index) can provide more comprehensive environmental monitoring. Edge AI can be strengthened to enable faster real-time decision-making even in low-connectivity remote areas. Incorporating GPS and geo-tagging features can help in location-based hazard tracking and emergency response. The system can also be extended with a mobile application and IoT cloud platforms for better user interaction and remote accessibility. Furthermore, implementing automatic model retraining using continuous incoming data can improve adaptability and long-term performance. Integration of renewable energy sources like solar power can make the system more sustainable for remote deployments. Enhancing cybersecurity measures will ensure data privacy and protection against potential threats. Finally, incorporating predictive maintenance features can help in early detection of sensor failures and system faults.

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